



O OBESO CRÍTICO

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Mini-Roteiro

- Epidemiologia e classificação
- Via aérea
- Ventilação mecânica
- ARDS
- Aparelho Urinário
- Aparelho Cardiovascular
- Imunidade e infecção
- Nutrição
- Alterações Farmacocinéticas



INTRODUÇÃO

- Incidência mundial triplicou desde 1975
- Prevalência de 25% na população de doentes na UTI

Elevada morbidade????

Elevada mortalidade????????



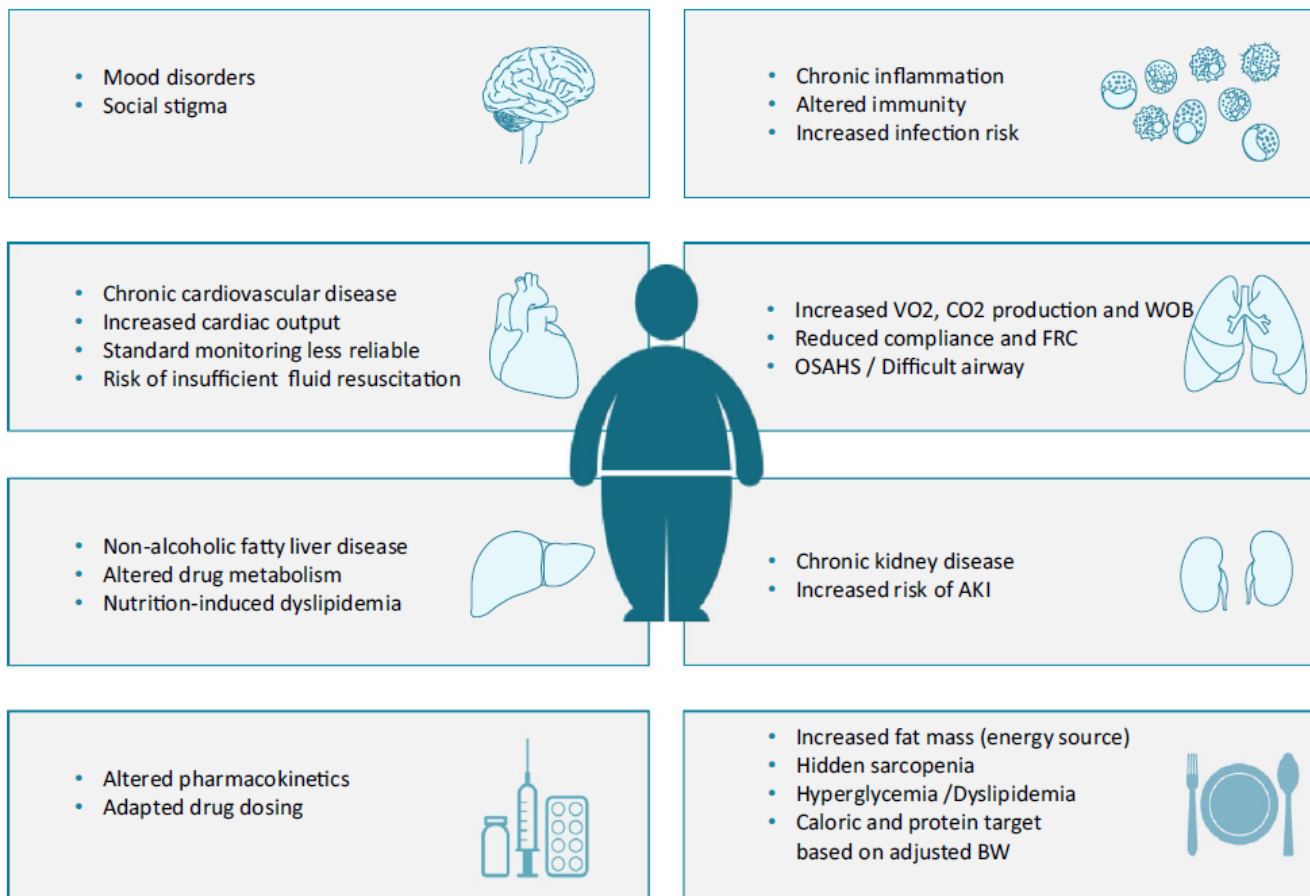


Fig. 1 Impact of obesity on organ systems and their management during critical illness. *WOB* work of breathing, *FRC* functional residual capacity, *OSAHS* obstructive sleep apnea and hypoventilation syndrome, *BW* body weight



Major depression (OR 1·21)
Bipolar disorder (OR 1·47)
Alzheimer's disease (RR 2·04); any dementia (RR 1·64)
Postoperative cognitive dysfunction (RR 1·27)

Hypertension (OR 4·8)
Heart failure (RR 1·90 M; RR 2·12 F)
Ischaemic heart disease (adjusted HR 1·64)
Myocardial infarction (adjusted HR 2·02)
Atrial fibrillation (adjusted HR 1·52 M; adjusted HR 1·46 F)

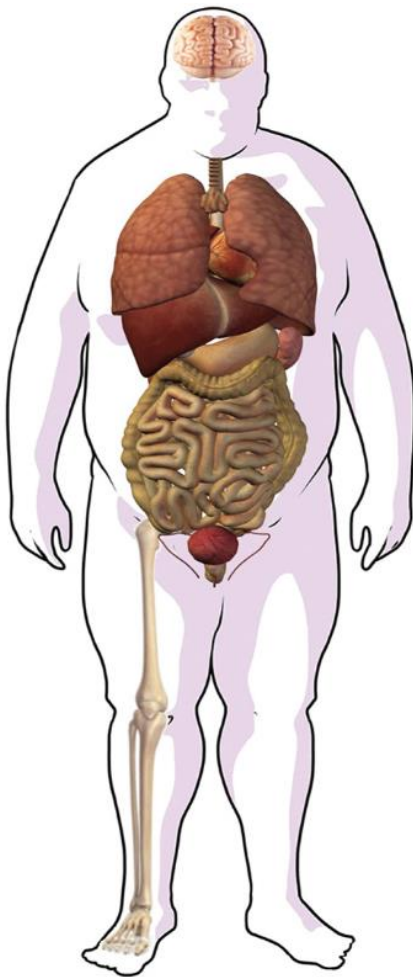
Diabetes mellitus (adjusted RR 7·28)
Dyslipidaemia (adjusted OR 2·2)

Metabolic syndrome
Stroke (OR 2·16)
Myocardial infarction (OR 2·01)
Atrial fibrillation (adjusted HR 1·52 M; adjusted HR 1·46 F)

Hypercoagulability
Stroke, myocardial infarction (OR 1·57)
Deep venous thrombosis (RR 2·50)
Pulmonary embolism (RR 2·21)

Non-alcoholic steatohepatitis (RR 4·6); cirrhosis (RR 4·1)

Renal disease (OR 1·38 hypertension; OR 1·4 type 2 diabetes)



Obstructive sleep apnoea (OR 6·0 for $\uparrow$ 10% bodyweight)
DMV (OR 3·39), DEI (OR 3·46) or both (OR 4·12)
Postoperative desaturation (OR 2·27)
Postoperative respiratory failure (OR 2·43)
Postoperative reintubation (OR 2·05)
Postoperative cardiac adverse events (OR 2·07)
Postoperative ICU transfer (OR 2·81)

Obesity hypoventilation syndrome
Cardiac event or cor pulmonale (OR 9)
Postoperative respiratory failure (OR 10·9)
Postoperative heart failure (OR 5·4)
Postoperative prolonged intubation (OR 3·1)
Postoperative ICU transfer (OR 10·9)

Respiratory disease
Respiratory complications with severe $\downarrow$ FEV1 (OR 2·97)
Cardiovascular complications with severe $\downarrow$ FEV1 (OR 2·02)

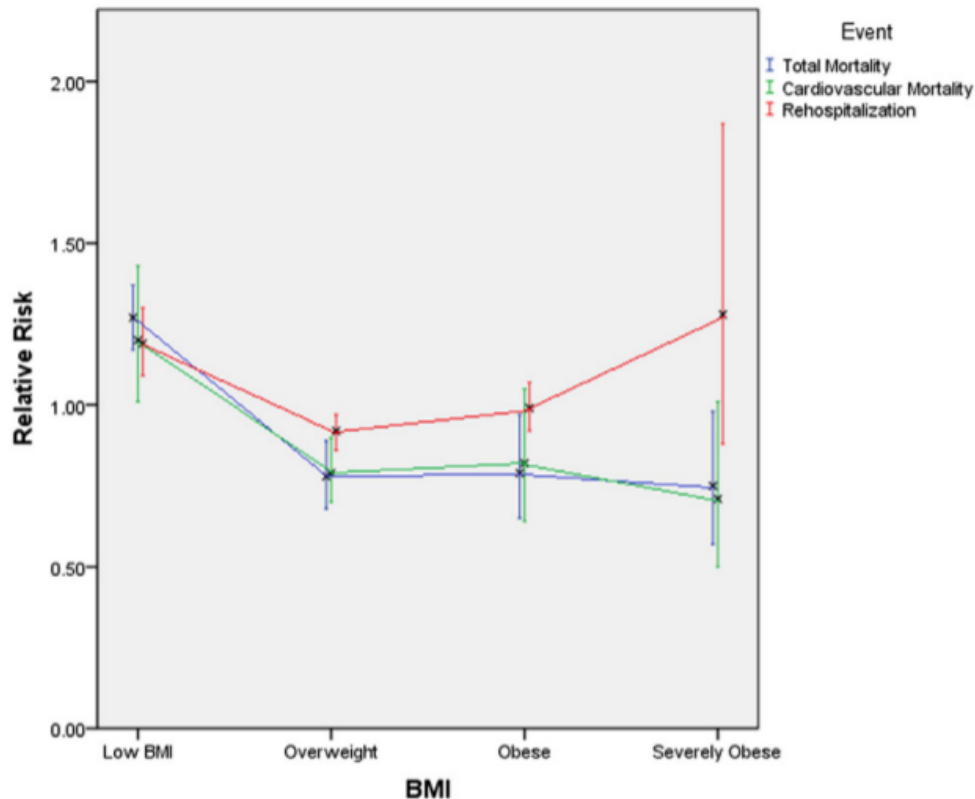
Asthma (RR 2·7)
Postoperative respiratory complications (OR 2·94)

Gastro-oesophageal reflux disease (OR 1·94)
Postoperative complications (OR 10·9)

Osteoarthritis (RR 1·12 hip; RR 1·25 knee)
Gout (RR 2·67)

Meta-Analysis of the Relation of Body Mass Index to All-Cause and Cardiovascular Mortality and Hospitalization in Patients With Chronic Heart Failure

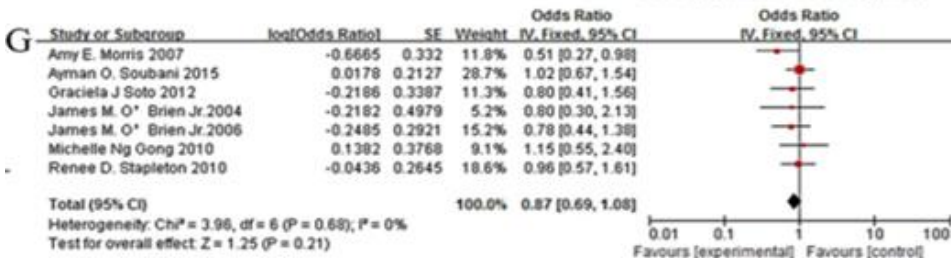
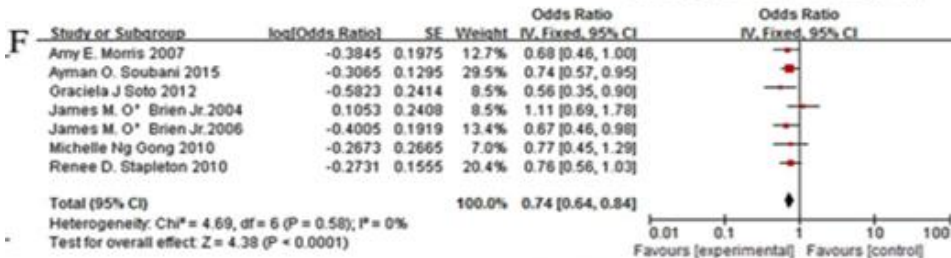
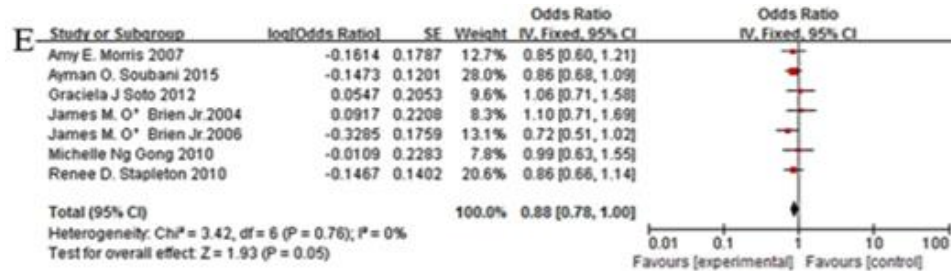
Abhishek Sharma, MD^{a,*}, Carl J. Lavie, MD^{b,c}, Jeffrey S. Borer, MD^{a,d}, Ajay Vallakati, MD^e



“Obesity Paradox” in Acute Respiratory Distress Syndrome: Asystematic Review and Meta-Analysis

Guo Zhi, Wang Xin, Wang Ying, Xing Guohong, Liu Shuying*

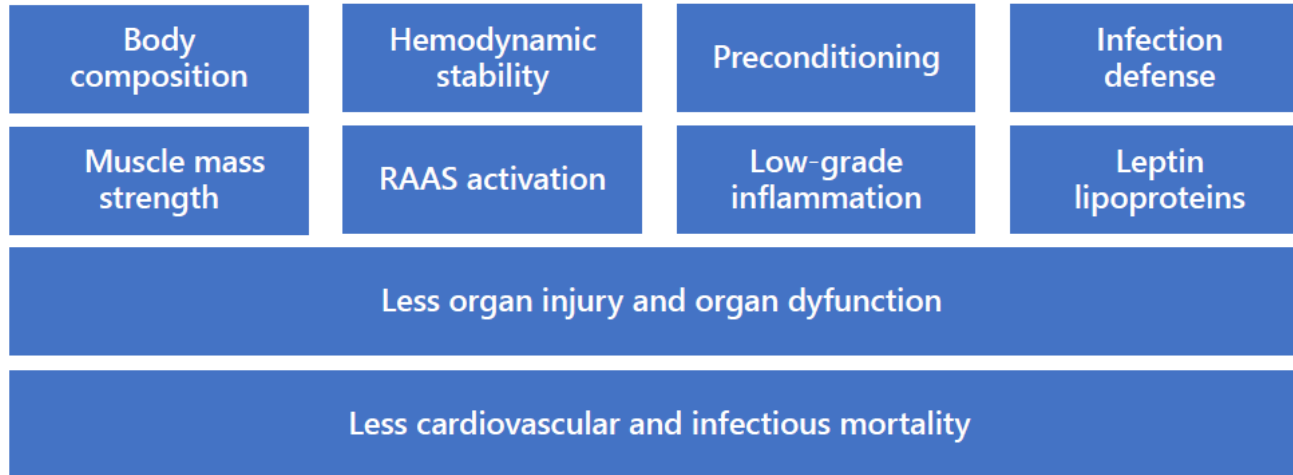
Department of Pulmonary Diseases, Jinan Military General Hospital, Jinan, 250031, China



Authors (year); (Ref. No.)	Exposure	Outcomes	Effect of Obesity (95% CL)	Heterogeneity*
Akinnusi <i>et al.</i> (2008); (2)	BMI ≥ 30 kg/m ² vs. < 30 kg/m ²	ICU mortality Hospital mortality Duration of mechanical ventilation	RR 1.0 (0.86, 1.16) RR 0.83 (0.74, 0.92) 1.48 (0.07, 2.89) more days	Substantial ($I^2 = 75.6\%$) Moderate ($I^2 = 56.4\%$) Moderate ($I^2 = 60.1\%$)
Hogue <i>et al.</i> (2009); (57)	BMI categories (kg/m ²): • Underweight (<18.5) • Normal weight (18.5–24.9) • Overweight (25–29.9) • Obese (30–39.9) • Morbidly obese (≥ 40)	ICU length of stay ICU mortality Hospital mortality Obese vs. normal Morbidly obese vs. normal Length of mechanical ventilation ICU length of stay Overweight or obese vs. normal Morbidly obese vs. normal Hospital length of stay Overweight or obese vs. normal Morbidly obese vs. normal	1.08 (0.27, 1.88) more days No significant differences for any weight category vs. normal [†] RR 0.76 (0.59, 0.92) RR 0.83 (0.66, 1.04) No significant differences for any weight category vs. normal [†] No significant difference [†] 0.77 (–0.14, 2.03) more days No significant difference [†] 1.36 (–2.08, 5.52) more days	High ($I^2 = 98.8\%$) Moderate–substantial ($I^2 = 50–80\%$) High (not specified) [‡] High ($I^2 = 93\%$) Low–substantial ($I^2 = 10–74\%$) Low ($I^2 < 40\%$) Low ($I^2 < 40\%$) Low ($I^2 < 13\%$) Low ($I^2 < 13\%$)
Oliveros and Villamor (2008); (123)	BMI categories (kg/m ²): • Underweight (<18.5) • Normal weight (18.5–24.9) • Overweight (25–29.9) • Obese (30–39.9) • Morbidly obese (≥ 40)	Mortality [§] Overweight vs. normal Obese vs. normal Morbidly obese vs. normal ICU length of stay Overweight vs. normal Obese vs. normal Morbidly obese vs. normal	OR 0.91 (0.84, 0.98) OR 0.82 (0.68, 0.98) OR 0.94 (0.82, 1.07) 0.17 (0.08, 0.26) more days 0.10 (–0.54, 0.74) more days 1.10 (0.84, 1.37) more days	Low ($I^2 = 29.6\%$) Substantial ($I^2 = 63.2\%$) Low ($I^2 = 24.0\%$) Low ($I^2 = 0\%$) Low ($I^2 = 0\%$) Low ($I^2 = 0\%$)



“Paradoxo do obeso crítico”



ANTROPOMETRIA

⇒ IMC vem sendo questionado como ferramenta para definição de obesidade = diferentes composições corporais / fenótipos (massa sólida não gordurosa, água corporal intra e extracelular e gordura subcutânea/visceral)

> Relação cintura-quadril, TC, absorbometria de energia dupla refletem melhor

➔ A maior parte da literatura, contudo, usa IMC para descrever os indivíduos



CLASSIFICAÇÃO OMS

DIAGNÓSTICO	IMC	RISCO ESTIMADO CA < 94 H < 80 M	RISCO ESTIMADO CA > 94 H > 80 M
ABAIXO DO PESO	< 18.5		
PESO NORMAL	18.5-24.9		
SOBREPESO	25-29.9	AUMENTADO	ALTO
OBESIDADE GI (OBESIDADE MODERADA)	30-34.9	ALTO	MUITO ALTO
OBESIDADE GII (OBESIDADE GRAVE)	35-39.9	MUITO ALTO	MUITO ALTO
OBESIDADE GIII (MORBIDA)	>= 40	EXTREMAMENTE ALTO	EXTREMAMENTE ALTO



ANTROPOMETRIA

Peso corporal total = actual / TBW

IBW (homem) = $51.65 \text{ kg} + 0.74 \text{ kg (H-150)}$

IBW (mulher) = $48.67 \text{ kg} + 0.66 \text{ kg (H-150)}$

ABW = $(\text{TBW} - \text{IBW}) \cdot 0.4 + \text{IBW}$ (calculado com 40% do excesso de peso) - IBW 120-200% TBW

Lean Body Weight = TBW – Body Fat Weight

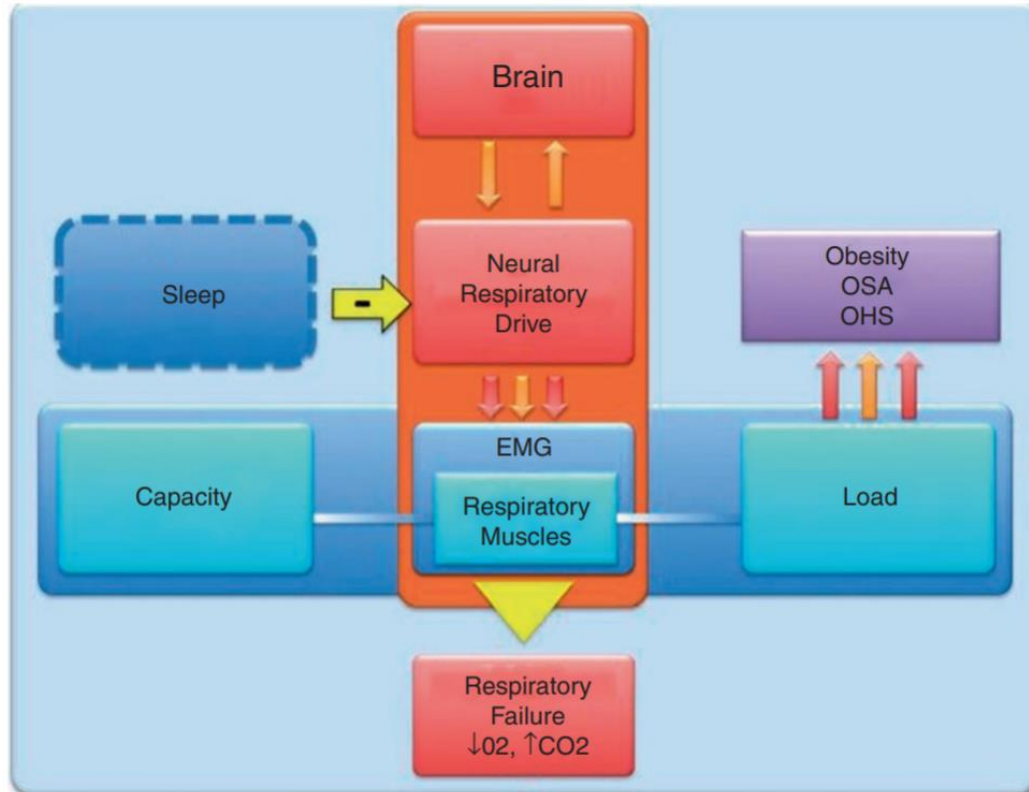




APARELHO PULMONAR



MECÂNICA RESPIRATÓRIA



MECÂNICA RESPIRATÓRIA

Aumento da PIA - sobrecarga à excursão do diafragma em posição supina

CONTRAPOSIÇÃO PALV / PIPL

CRF EM VALOR PRÓXIMO A VR

AUMENTO RAW + PEEPi

BAIXO GRADIENTE VA DISTAIS > COLAPSO BRONQUIOLAR

AUMENTO WOB → DRIVE NEURAL

RNC

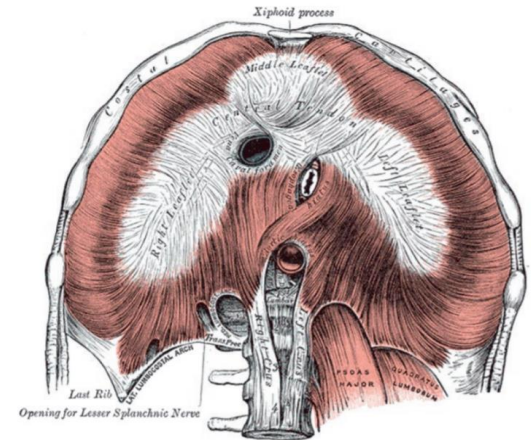


Fig. 2.1 The human diaphragm (reproduced from Gray's Anatomy, 20th US edition, originally published in 1918)

FADIGA





The Effects of Body Mass Index on Lung Volumes*

Richard L. Jones, PhD, FCCP; and Mary-Magdalene U. Nzekwu, BSc

Table 1—Pulmonary Function Results for the Different BMI Groups*

Variables	BMI Groups, kg/m ²				
	20–25	25–30	30–35	35–40	> 40
Gender					
Female	58	38	50	41	28
Male	35	40	42	26	15
FEV ₁ /FVC ratio	100.7 (5.6)	101.4 (5.5)	102.8 (4.9)	102.2 (5.9)	101.5 (4.2)
VC	97.6 (10.0)	97.2 (10.5)	92.2 (11.2)	91.1 (10.9)	87.9 (11.4)
TLC	98.7 (8.7)	96.9 (9.0)	93.1 (9.1)	92.0 (8.3)	88.1 (10.7)
RV	102.7 (15.9)	96.7 (18.3)	95.5 (16.4)	94.6 (16.3)	90.5 (17.2)
RV/TLC	29.6 (7.2)	30.5 (7.9)	32.1 (8.5)	32.6 (8.2)	31.0 (6.2)
FRC	103.1 (15.5)	89.2 (14.1)	78.3 (13.1)	72.2 (12.9)	66.6 (12.3)
FRC/TLC	52.9† (6.5)	46.2‡ (6.1)	42.6 (6.0)	44.4 (5.3)	42.2 (6.0)
ERV	95.6 (29.8)	72.3 (30.6)	42.4 (29.3)	29.3 (18.7)	24.6 (18.8)
DLCO	99.3 (10.9)	101.3 (10.7)	101.2 (11.3)	103.2 (12.4)	108.1† (8.5)

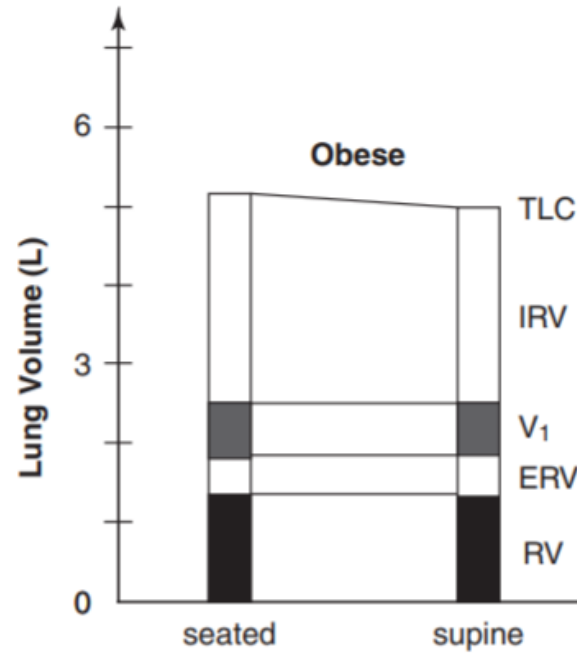
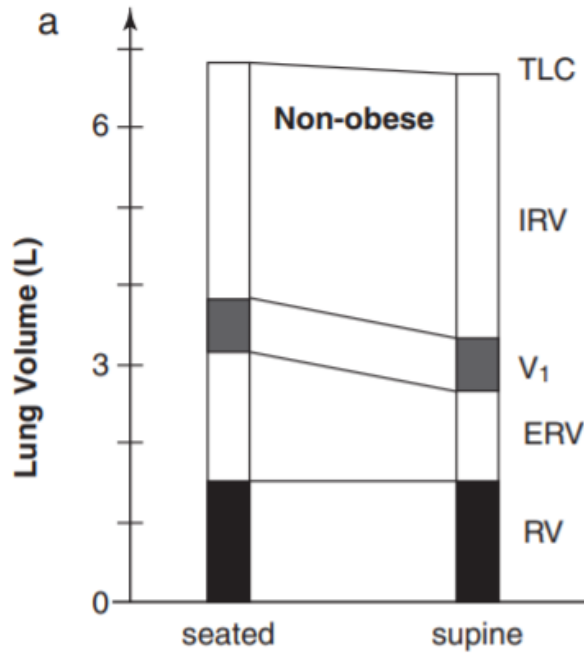
*Values are expressed as No. or mean percentage of predicted (SD), except for RV/TLC and FRC/TLC, which are absolute ratios. Comparisons between BMI groups for VC, TLC, RV, FRC, and ERV are illustrated in Figures 1, 2.

†For RV/TLC, FRC/TLC, and DLCO, $p < 0.05$ compared to all other BMI groups.

‡For RV/TLC, FRC/TLC, and DLCO, $p < 0.05$ compared to adjacent BMI groups.



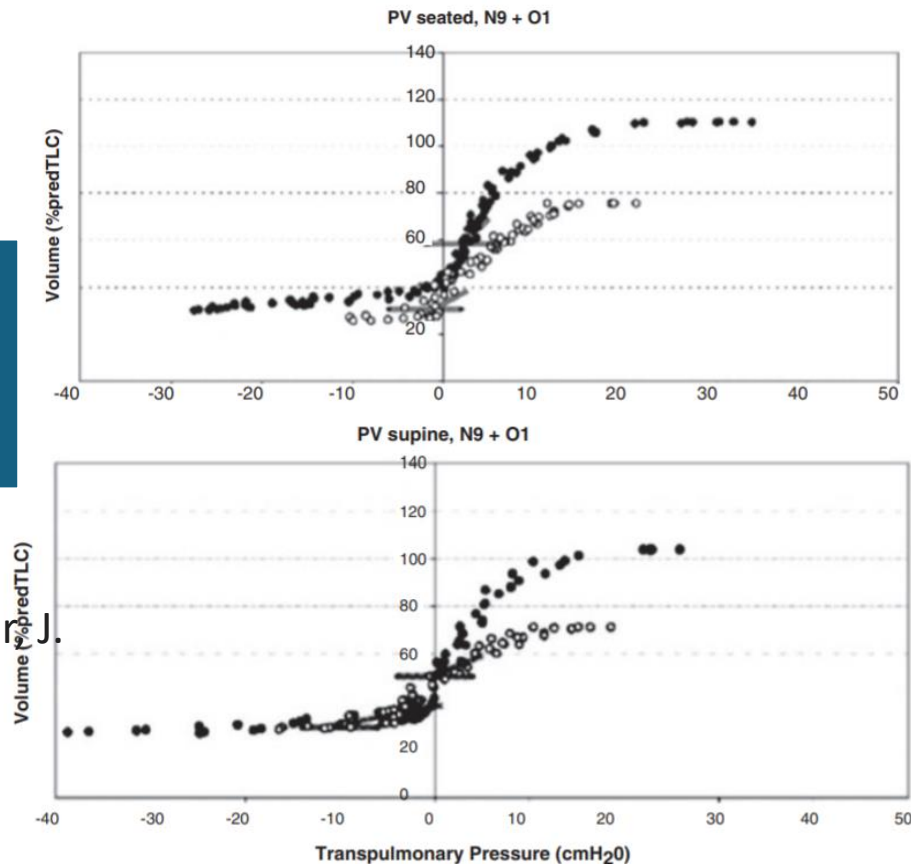
VOLUMES PULMONARES



1. Homem, 66 anos, IMC = 23.3, Altura 1.72m

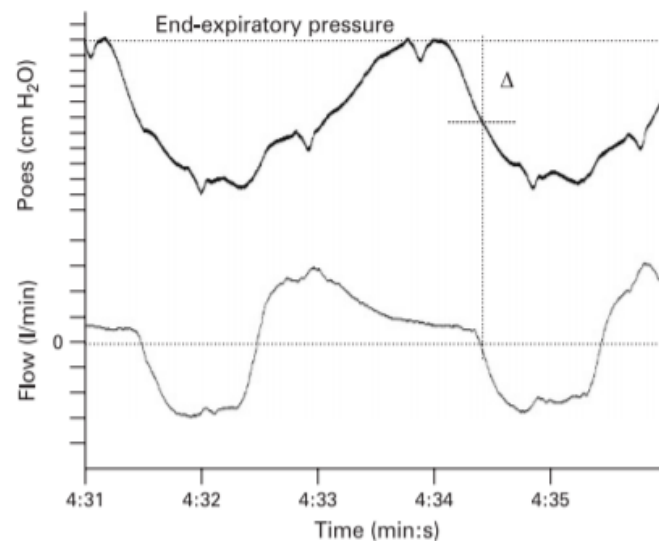
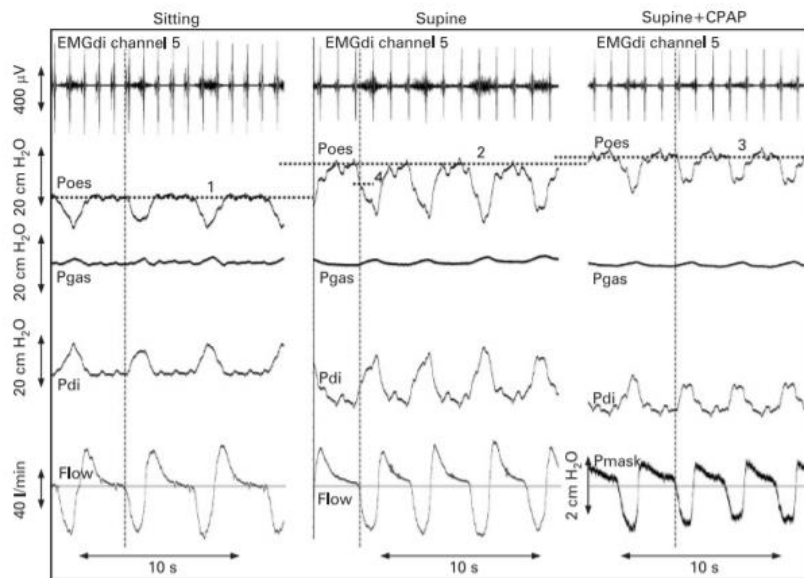
2. Homem, 60 anos, 1.73m, IMC = 34.4

Ratneswaran, C., Murphy, P., Hart, N., Steier J.
(2018)



Neural respiratory drive in obesity

J Steier,^{1,2} C J Jolley,^{1,2} J Seymour,^{1,2} M Roughton,^{2,3} M I Polkey,² J Moxham¹



Paciente IMC 42 kg/m²
Circunferência de pescoço 43 cm
PEEPi = 6 cm H₂O

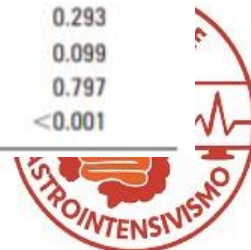


Neural respiratory drive in obesity

J Steier,^{1,2} C J Jolley,^{1,2} J Seymour,^{1,2} M Roughton,^{2,3} M I Polkey,² J Moxham¹

Table 2 Electromyography (EMG) and respiratory variables during resting breathing with mouthpiece when seated and supine

	Obese group, seated	Non-obese group, seated	p Value	Obese group, supine	Non-obese group, supine	p Value
EMGdi (%max)	21.9 (9.0)	8.4 (4.0)	<0.001	24.7 (8.2)**	7.0 (3.4)	<0.001
EMGneck (%max)	5.6 (6.3)	2.0 (1.7)	0.004	6.8 (10.7)	2.6 (1.6)	0.036
EMGpara (%max)	7.7 (4.3)	4.4 (3.0)	0.001	9.7 (9.2)	5.0 (2.9)	0.010
EMGabdominal (%max)	3.3 (3.1)	1.6 (1.2)	0.005	4.0 (2.8)	1.5 (1.4)	<0.001
Pressure swing, Poes (cm H ₂ O)	9.6 (2.9)	5.3 (2.2)	<0.001	16.0 (5.0)***	6.9 (2.0)***	<0.001
Pressure swing, Pgas (cm H ₂ O)	5.0 (2.3)	4.9 (1.7)	0.932	3.2 (1.4)***	3.5 (1.3)***	0.371
Pressure swing, Pdi (cm H ₂ O)	12.9 (3.8)	8.9 (2.3)	<0.001	18.2 (4.5)***	9.7 (2.3)*	<0.001
End-expiratory baseline pressure, Poes (cm H ₂ O)	4.0 (4.2)	-2.0 (2.7)	<0.001	12.8 (4.8)***	5.1 (3.5)***	<0.001
End-expiratory baseline pressure, Pgas (cm H ₂ O)	15.8 (5.5)	10.9 (5.7)	0.001	13.0 (7.8)*	7.8 (3.5)**	<0.001
Respiratory rate (breaths/min)	16.4 (3.1)	16.0 (4.5)	0.693	16.9 (4.2)	15.7 (4.5)	0.293
Minute ventilation (l/min)	10.1 (3.0)	8.9 (1.7)	0.060	10.0 (2.5)	8.9 (2.7)	0.099
V _T (l)	0.64 (0.26)	0.61 (0.20)	0.580	0.62 (0.22)	0.61 (0.24)	0.797
PEEPi (cm H ₂ O)	1.4 (1.5)	0.5 (0.3)	0.001	5.3 (3.6)***	0.7 (0.5)	<0.001



The Impact of Morbid Obesity on Oxygen Cost of Breathing ($\dot{V}O_{2\text{RESP}}$) at Rest

JOHN P. KRESS, ANNE S. POHLMAN, JOHN ALVERDY, and JESSE B. HALL

Department of Medicine, University of Chicago, Chicago, Illinois

Pacientes candidatos a by-pass gástrico pré e pós VM

Aumento do consumo de oxigênio – aumento da sobrecarga poderia saturar mecanismo?

$\dot{V}O_2$ basal = 354.6 versus 221.4 ml/min; $p = 0.0001$

DeltaObeso x DeltaControle

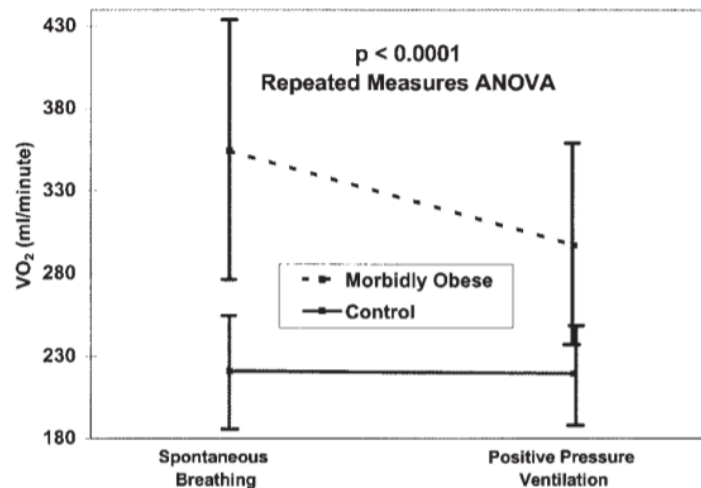


Figure 1. Mean ($\pm$ SD) $\dot{V}O_2$ in morbidly obese and control patients—spontaneous breathing versus positive pressure ventilation.



Diaphragm Activity in Obesity

RUY V. LOURENÇO

*From the Department of Medicine, University of Illinois College of Medicine
and The Hektoen Institute for Medical Research. Chicago. Illinois 60680*

TABLE II

*Arterial Blood Gases, Total Chest Compliance, and Ventilatory and Diaphragmatic Responses
to Carbon Dioxide Breathing in Eight Obese Subjects*

Subject	Arterial blood*			Chest compliance	Ventilatory responses	Diaphrag- matic re- sponses
	Oxygen tension	Carbon dioxide tension	pH			
	mm Hg	mm Hg		liters/cm H ₂ O	liters/min/mm Hg	mv-sec/min/ mm Hg
Group I Obese-Normal						
M. A.	83	41	7.39	0.092	1.70	90
S. C.	89	36	7.42	0.135	1.92	72
J. N.	76	42	7.41		2.13	48
M. F.	82	38	7.40		1.85	52
Average	83	39	7.40	0.114	1.90	66
Group II Obese-Hypoventilation						
G. G.	67	47	7.38	0.123	1.25	18
M. S.	44	51	7.35	0.056	0.91	22
L. D.	49	60	7.30	0.103	0.66	14
R. F.	65	55	7.37	0.084	0.82	12
Average	56	53	7.35	0.092	0.91	17

* Sampled at rest during room air breathing

8 doentes obesos = 4
com gasometria
normal + 4 com
hipercapnia e
hipoxemia



DRIVE NEURAL NA APNEIA OBSTRUTIVA



Increased load on the respiratory muscles in obstructive sleep apnea

Joerg Steier^{a,b,*}, Caroline J. Jolley^a, John Seymour^a, Katie Ward^a, Yuan M. Luo^c,

- 21 pacientes com SAHOS e 21 controles submetidos a eletromiograma de diafragma, submandibular / musculatura acessória

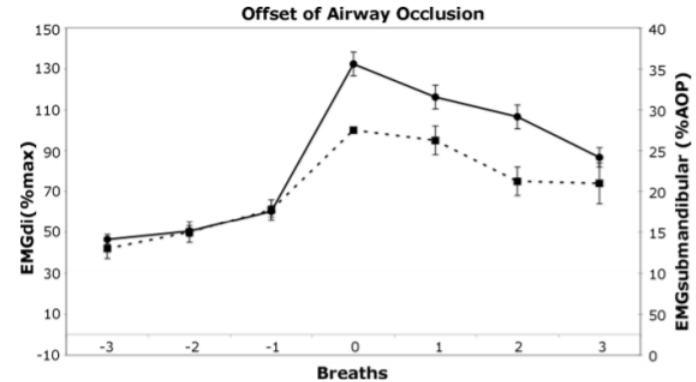
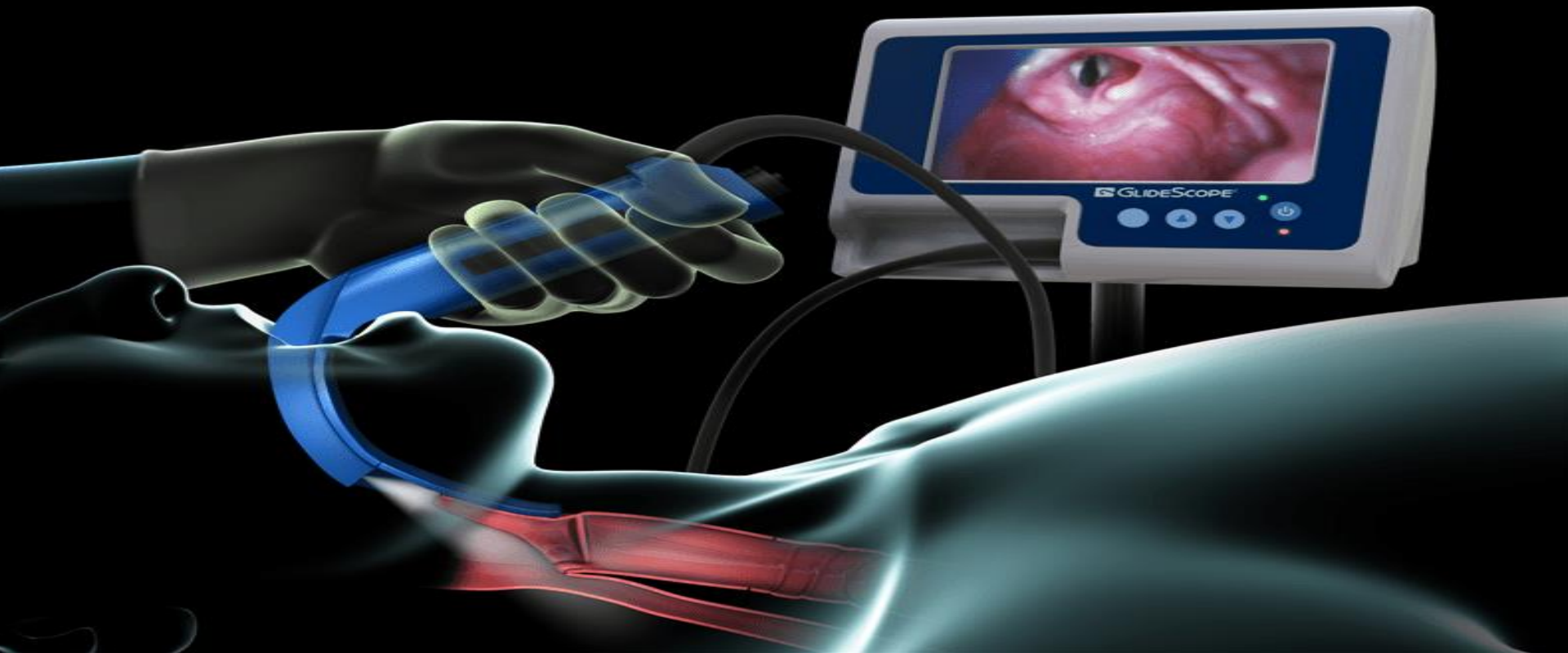


Table 3

EMG activity in unobstructed breathing awake and asleep. Mean (SD), sleep in stage N2, supine, all parameters in %max.

	EMG during unobstructed breathing in sleep		
	OSA patients	Normal subjects	p-Value
EMG _{di} , awake	13.1 (5.6)	8.1 (2.3)	<0.01
EMG _{di} , asleep	22.8 (19.2)	7.7 (3.9)	<0.001
EMG _{neck}	9.2 (10.7)	2.6 (1.8)	<0.001
EMG _{para}	12.5 (12.9)	4.1 (2.0)	<0.001
EMG _{abdomen}	6.7 (9.0)	1.5 (1.3)	<0.001



VIA AÉREA



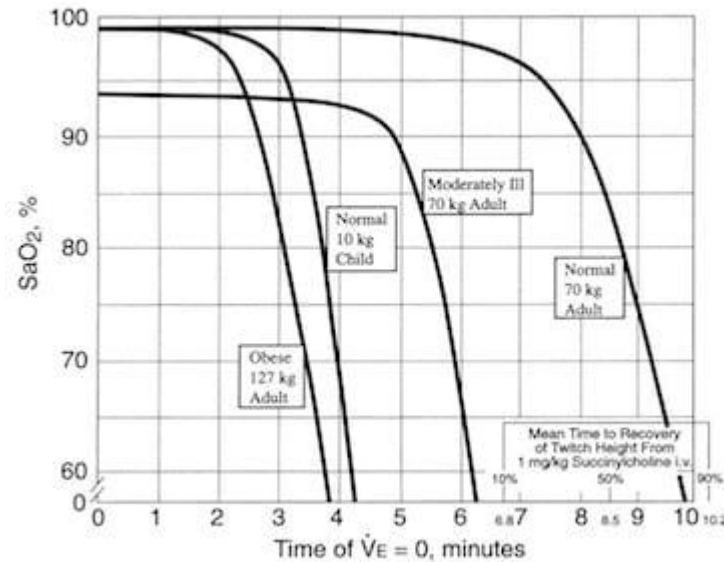
FATORES ASSOCIADOS A IOT DIFÍCIL

- Mallampati elevado
- Redução da abertura comissural
- Redução de mobilidade cervical
- Estados comatosos
- Hipoxemia severa
- Gordura facial e selamento de BVM



PRÉ-OXIGENAÇÃO

TIME TO HEMOGLOBIN DESATURATION WITH INITIAL $F_{A}O_2 = 0.87$



Anesth, 2005.



High-flow nasal cannula for pre- and apneic oxygenation during rapid sequence induction intubation in emergency surgery: A systematic review and meta-analysis

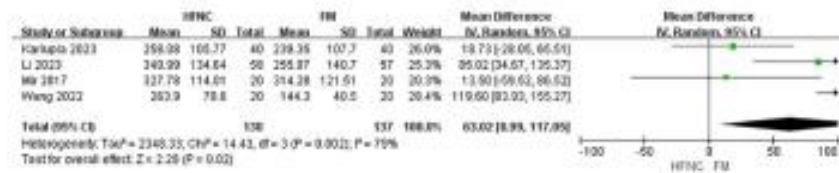


Fig 6. Forest plot of desaturation.

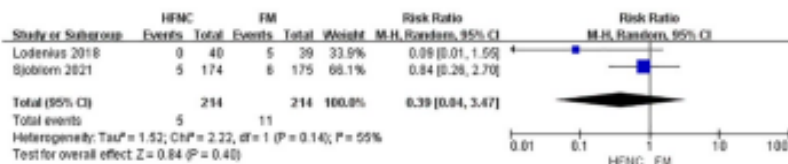
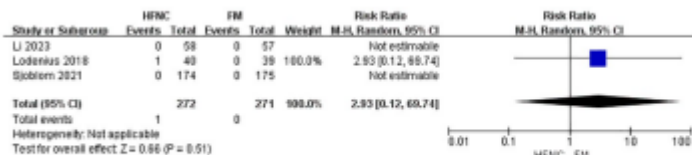


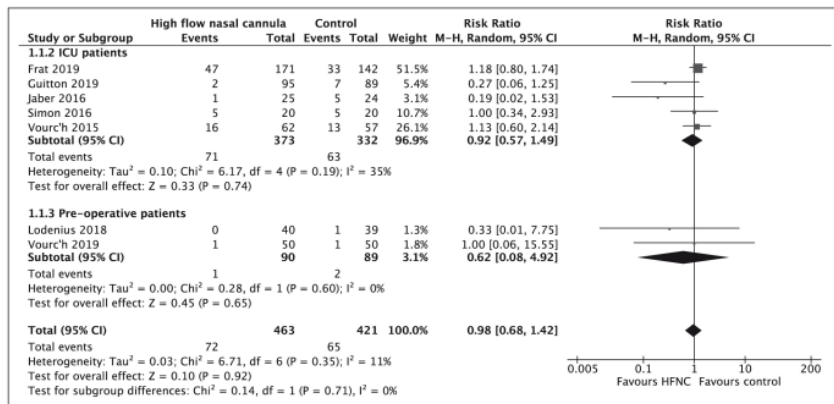
Fig 8. Forest plot of regurgitant aspiration.



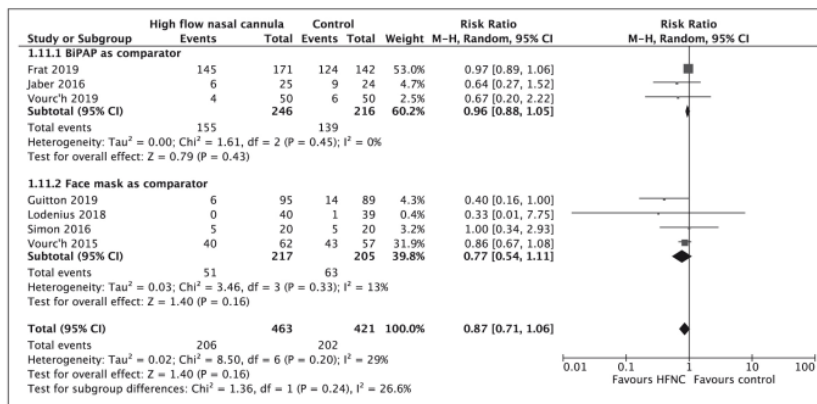
Moderate Certainty Evidence Suggests the Use of High-Flow Nasal Cannula Does Not Decrease Hypoxia When Compared With Conventional Oxygen Therapy in the Peri-Intubation Period: Results of a Systematic Review and Meta-Analysis

Dhruv Chaudhuri MD, David Granton BSc, Dominic Viana Wang BSc, Charan Eissa MD, MSc

Hipoxemia



Complicações pós IOT



Chaudhuri D et al. Crit Care Med. 2020 Apr;48(4):571-578.



The Effectiveness of Noninvasive Positive Pressure Ventilation to Enhance Preoxygenation in Morbidly Obese Patients: A Randomized Controlled Study

Delay, Jean-Marc, MD^{*}; Sebbane, Mustapha, MD^{*}; Jung, Boris, MD^{*}; Nocca, David, MD[†]; Verzilli, Daniel, MD^{*};

- 28 pacientes submetidos a procedimentos cirúrgicos
- Endpoint: avaliação número de pacientes com ETO₂ > 95% após 5 min

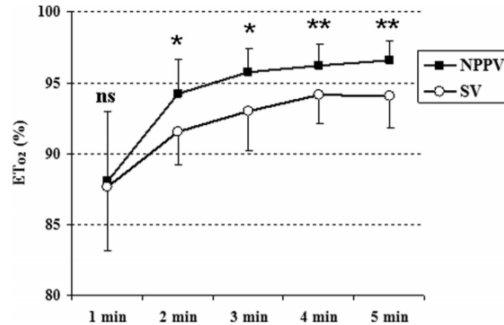
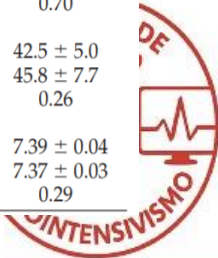


Table 2. Comparison of Blood Gas Results for Patients Randomly Assigned with the SV and NPPV Groups

	Before preoxygenation	After 5 min of preoxygenation	At Spo ₂ = 95%	After 5 min of ACV
Pao ₂ (mm Hg)				
SV	86.0 ± 9.3	454.3 ± 93.4	66.1 ± 7.8	130.8 ± 44.3
NPPV	87.6 ± 11.8	507.1 ± 81.0	69.2 ± 11.1	140.6 ± 47.3
P	0.93	0.12	0.56	0.67
Sao ₂ (%)				
SV	96.4 ± 1.5	99.9 ± 0.3	89.6 ± 3.4	97.6 ± 3.2
NPPV	96.7 ± 1.3	99.9 ± 0.3	90.2 ± 4.9	97.0 ± 4.0
P	0.93	0.99	0.49	0.70
Paco ₂ (mm Hg)				
SV	37.9 ± 4.8	41.7 ± 5.2	55.3 ± 7.6	42.5 ± 5.0
NPPV	39.4 ± 5.6	38.2 ± 7.0	55.3 ± 7.4	45.8 ± 7.7
P	0.69	0.17	0.69	0.26
pH				
SV	7.42 ± 0.04	7.40 ± 0.03	7.33 ± 0.04	7.39 ± 0.04
NPPV	7.41 ± 0.02	7.43 ± 0.04	7.32 ± 0.03	7.37 ± 0.03
P	0.44	0.08	0.77	0.29



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Valérie Gissot
Stéphanie Ragot
Arnaud Desachy
Isabelle Runge
Christine Lebert
René Robert
for the Association des
Réanimateurs
du Centre-Ouest (ARCO)
study group

Impact of obesity in mechanically ventilated patients: a prospective study

Table 2 ICU course in obese and nonobese patients

Complication	Obese (<i>n</i> = 82)	Nonobese (<i>n</i> = 124)
Acquired acute renal failure: no. (%)	11 (13.4%)	11 (8.9%)
Renal replacement therapy: no. (%)	9 (10.9%)	16 (12.9%)
Pressure ulcers: no. (%)	12 (15%)	20 (16%)
Ventilator-associated pneumonia per 1,000 days: mean [SD]	7.8 [2.6]	12.4 [2.5]
Duration of catheterization (days): mean [SD]	9 [6]	10 [8]
Number catheter per 1,000 patient-ICU days	71	61
Infection per 1,000 catheter-days: mean [SD]	9.5 [3.6]	7.7 [2.6]
Post-extubation stridor: no. (%)	10/65 (15.3%)	3/90 (3.3%)*

SD Standard deviation

* *P* = 0.008



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Table 3 Univariate analysis: factors of mortality

Variable	ICU mortality			Hospital mortality		
	Alive (n = 155)	Dead (n = 51)	P value	Alive (n = 143)	Dead (n = 63)	P value
Age (years): mean [SD]	65 [12]	65 [10]	0.73	65 [12]	65 [10]	0.79
Men/women: no	94/61	35/16	0.31	86/57	43/20	0.26
Obese patients: no. (%)	62 (40%)	20 (39%)	0.92	58(41%)	24 (38%)	0.74
Ventilator-days (days): median (IQR)	10 (6–18)	12 (7–23)	0.28	10 (6–17)	13 (7–23)	0.12
Primary diagnosis: no. (%)			0.06			0.11
Acute exacerbations of chronic respiratory failure	57 (37%)	10 (20%)		54 (38%)	13 (21%)	
Acute hypoxic respiratory failure	39 (25%)	12 (23%)		34 (24%)	17 (27%)	
Airway protection-related neurologic disease	23 (15%)	9 (18%)		20 (14%)	12 (19%)	
Shock	36 (23%)	20 (39%)		35 (24%)	21 (33%)	
Renal replacement therapy: no. (%)	11 (7%)	14 (27%)	0.0001	10 (7%)	15 (24%)	0.0007
SAPS II score: mean [SD]	43 [14]	52 [15]	0.0002	42 [13]	52 [15]	<0.0001
PaO ₂ /FiO ₂ day 1: mean [SD]	213 [117]	194 [115]	0.32	210 [116]	206 [119]	0.81
PaO ₂ /FiO ₂ day 2: mean [SD]	226 [103]	200 [114]	0.14	226 [103]	206 [112]	0.24
Difficulty of intubation: no. (%)	12 (8%)	8 (16%)	0.11	10 (7%)	10 (16%)	0.05
Transfer from ward: no (%)	95 (71%)	36 (61%)	0.23	86 (60%)	45 (71%)	0.12

SAPS II Simplified acute physiology score, IQR interquartile range, SD standard deviation, ICU intensive care unit



Difficult Tracheal Intubation Is More Common in Obese Than in Lean Patients

Philippe Juvin, MD, PhD, Elisabeth Lavaut, MD, Hervé Dupont, MD, Pascale Lefevre, MD, Monique Demetriou, MD, Jean-Louis Dumoulin, MD, and Jean-Marie Desmonts, MD

IDS>5 EM 20 OBESOS VERSUS 3 MAGROS (p = 0.0001)

Table 3. Multivariate Analysis Performed in the Obese Patients to Determine the Independent Risk Factors for Difficult Intubation in This Population

Variable	β	SD	P value	Odds ratio	95% CI	
					Lower	Upper
Mallampati class III–IV	2.53	0.93	0.007	12.51	2.01	77.81
Constant	−4.69	0.97	<0.001			



High Body Mass Index Is a Weak Predictor for Difficult and Failed Tracheal Intubation

A Cohort Study of 91,332 Consecutive Patients Scheduled for Direct Laryngoscopy Registered in the Danish Anesthesia Database

Lars H. Lundstrøm, M.D.,* Ann M. Møller, M.D., Dr.Med.Sci.,† Charlotte Rosenstock, M.D., Ph.D.,‡ Grethe Astrup, M.D.,§ Jørn Wetterslev, M.D., Ph.D.||

Table 4. Multivariate Model for the Prediction of Difficult Intubation

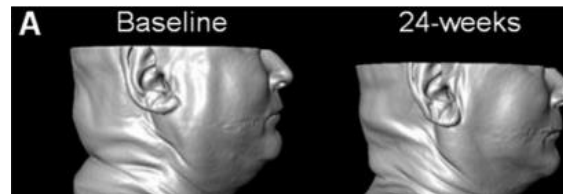
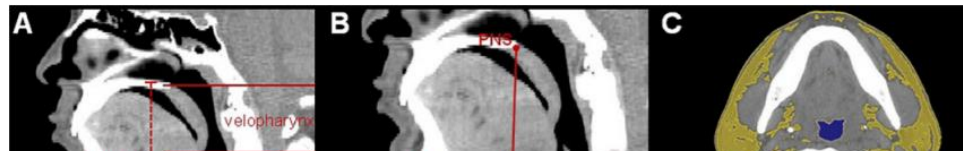
	OR	95 % CI	P
35 ≤ BMI	1.34	1.19–1.51	< 0.0001
25 ≤ BMI < 35	1.11	1.04–1.18	< 0.0016
Mallampati III and IV	3.70	3.41–4.00	< 0.0001
Male	1.35	1.27–1.44	< 0.0001
Age, year	1.01	1.01–1.01	< 0.0001
Surgical Priority = Scheduled	1.34	1.24–1.44	< 0.0001
Previous difficult intubation = Yes	6.32	5.11–7.84	< 0.0001
Previous difficult intubation = Unknown	1.26	1.15–1.39	< 0.0001
NMBA = No relaxation	1.59	1.49–1.70	< 0.0001

Table 6. Accuracy of the Prediction of DTI with Dichotomous Stand-alone Tests of BMI and Weight (kg)

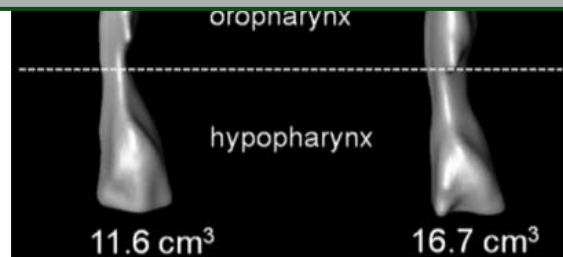
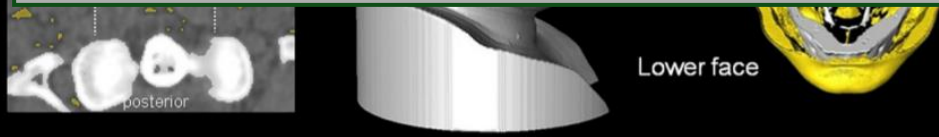
	TP	FP	TN	FN	OR	Sens.	Spec.	PPV	NPV	PosLR	NegLR
BMI ≥ 25	2526	41,368	45,249	2179	1.27	0.54	0.52	0.06	0.95	1.12	0.89
BMI ≥ 35	352	5136	81,481	4352	1.28	0.07	0.94	0.06	0.95	1.26	0.98
Weight ≥ 90	1121	17,159	69,458	3583	1.27	0.24	0.80	0.06	0.95	1.20	0.95
Weight ≥ 110	284	3942	82,675	4421	1.35	0.06	0.95	0.07	0.95	1.33	0.98



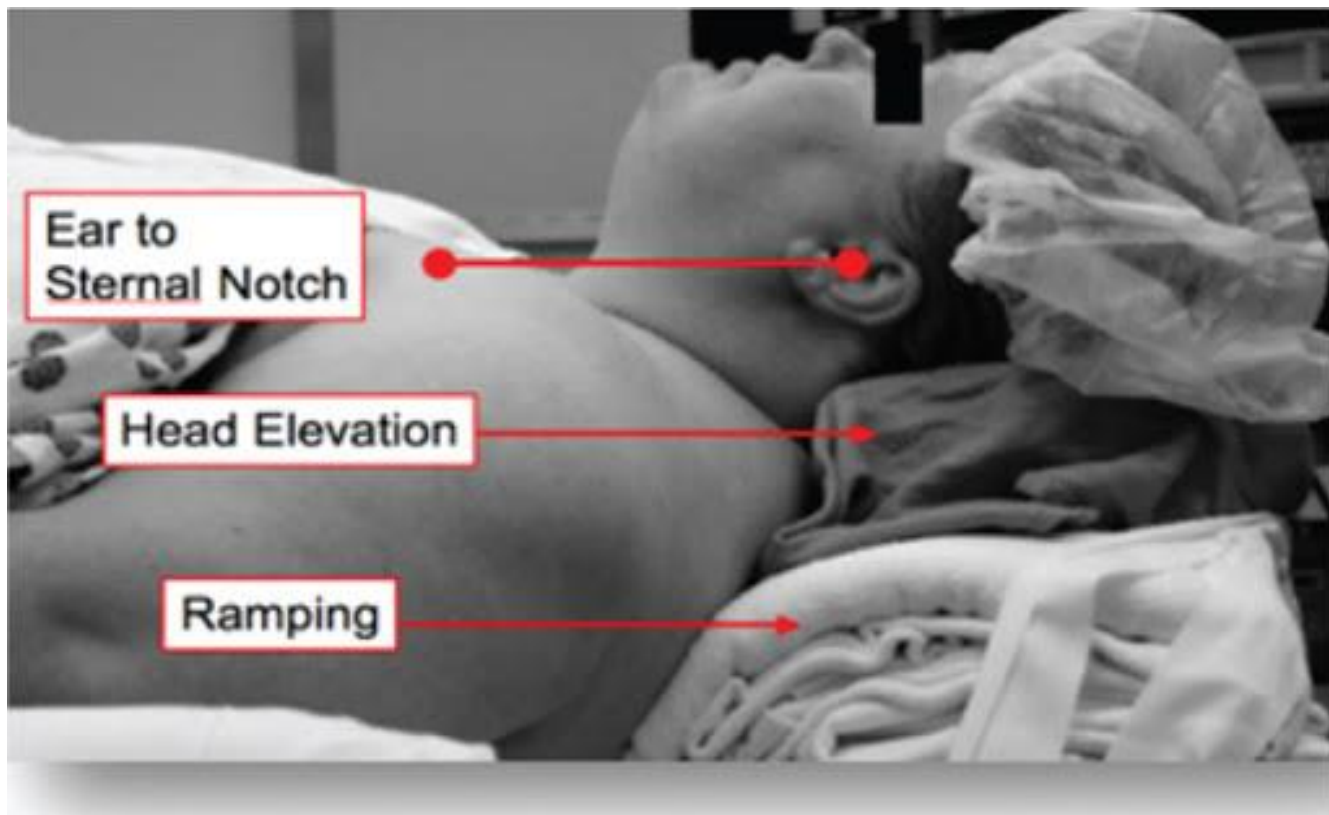
SAHOS E VIA AÉREA

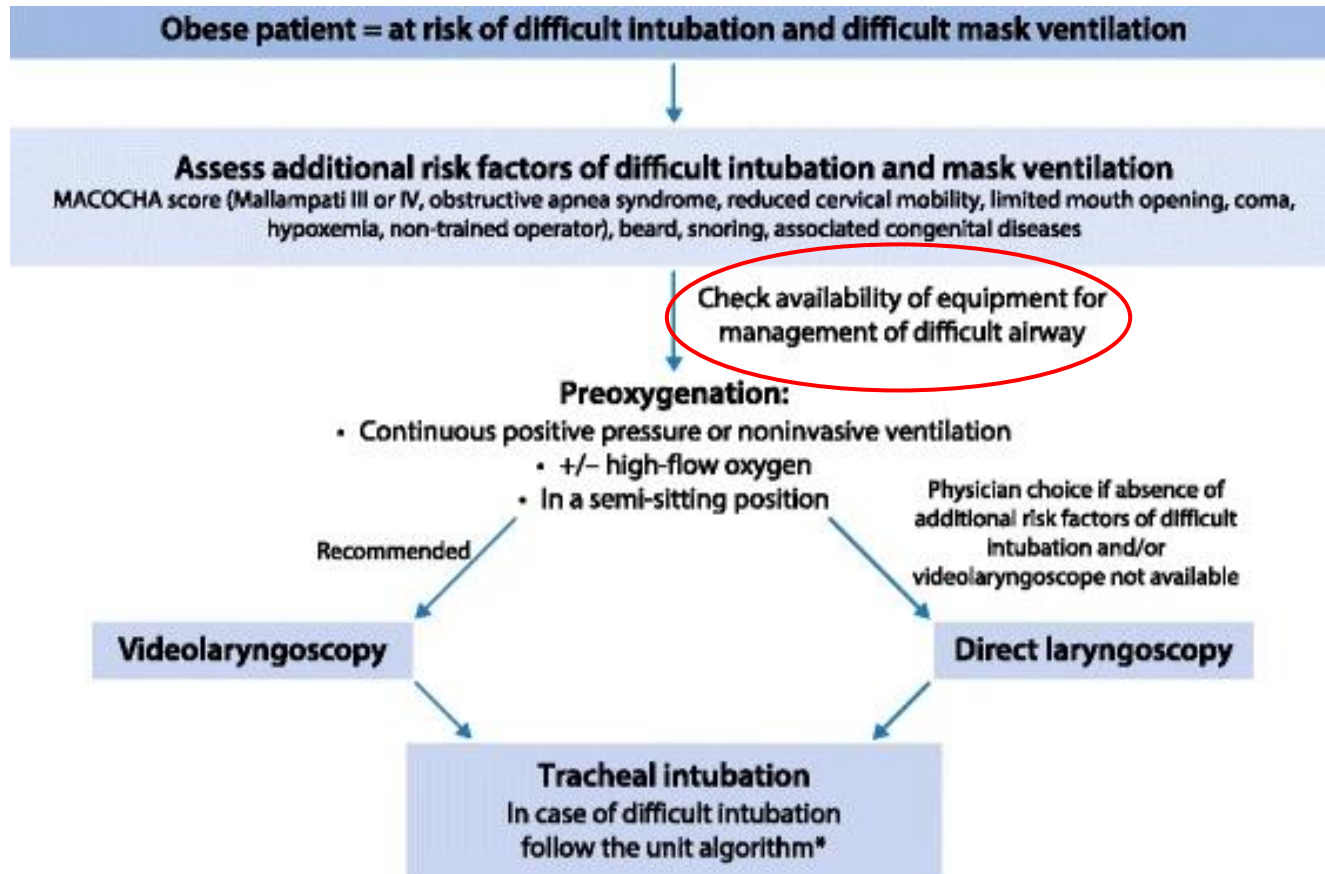


REVERSÍVEL COM PERDA
PONDERAL!!!



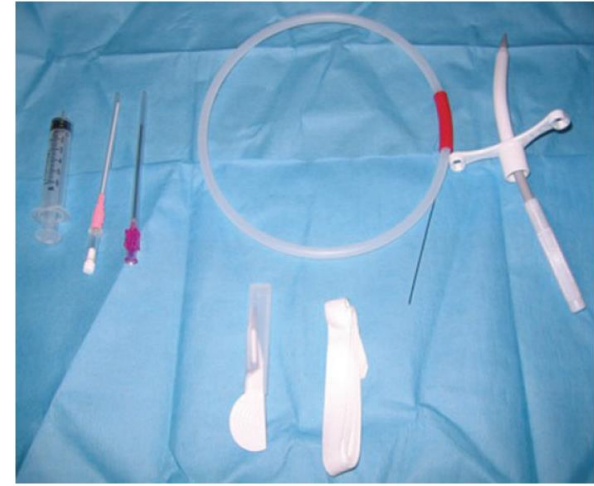
E O MANEJO...?





VIA AÉREA AVANÇADA

- ❖ Traqueostomia cirúrgica como método de escolha em obesos mórbidos? Traqueo precoce?
- ❖ Cânulas padrão curtas - uso de borda ajustável
- ❖ Maior incidência de deslocamento e malposicionamento
- ❖ Evitar crico por punção





VENTILAÇÃO NÃO INVASIVA



Noninvasive Ventilation in Acute Hypercapnic Respiratory Failure Caused by Obesity Hypoventilation Syndrome and Chronic Obstructive Pulmonary Disease

Andres Carrillo¹, Miquel Ferrer^{2,3}, Gumersindo Gonzalez-Diaz¹, Antonia Lopez-Martinez¹, Noemi Llamas¹, Maravillas Alcazar¹, Lucia Capilla¹, and Antoni Torres^{2,3}

TABLE 4. PATIENT OUTCOME*

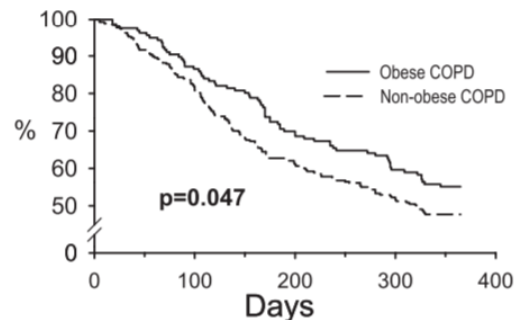
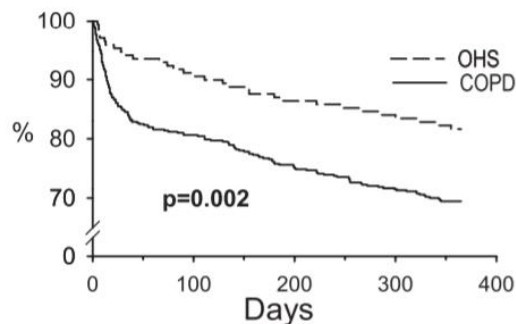
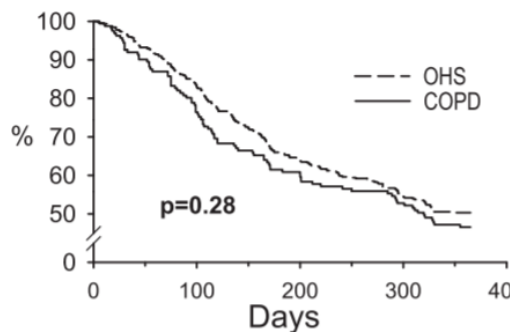
	COPD (n = 543)	OHS (n = 173)	P Value
NIV failure, n (%)	59 (11)	11 (6)	0.11
Endotracheal intubation, n (%)	23 (4)	7 (4)	<0.99
Time from onset of NIV to ETI, d	1.8 ± 1.4	2.7 ± 4.5	0.61
Late NIV failure, n (%)	67 (13)	12 (7)	0.037
Readmission to the ICU, n (%)	24 (5)	1 (1)	0.023
Maximal SOFA during ICU stay	5.2 ± 3.1	5.1 ± 2.7	0.64
ICU stay, d	4 ± 8	4 ± 4	0.93
Hospital stay, d	15 ± 13	16 ± 10	0.32
PaCO ₂ at hospital discharge, mm Hg	58 ± 6	53 ± 4	<0.001
ICU mortality, n (%)	42 (8)	2 (1)	0.003
Hospital mortality, n (%)	96 (18)	10 (6)	<0.001
Causes of hospital death			0.095
Respiratory failure [†]	76 (79)	5 (50)	
Shock-multiple organ failure	20 (21)	5 (50)	
Respiratory therapies during the year after the hospital admission [‡]			
Domiciliary nocturnal nasal CPAP, n (%)	18 (4)	71 (45)	<0.001
Domiciliary NIV, n (%)	52 (12)	16 (10)	0.66

- 716 pacientes, unicêntrico, Barcelona
- IRPA hipercápnica com pH < 7.43 e PCO₂ > 45 mmHg
- IPAPi 12 cmH₂O - aumento 2-3 cmH₂O até 25 / EPAPi 5 cmH₂O e aumentos 1-2 cmH₂O para hipoxemia e desconforto



Noninvasive Ventilation in Acute Hypercapnic Respiratory Failure Caused by Obesity Hypoventilation Syndrome and Chronic Obstructive Pulmonary Disease

Andres Carrillo¹, Miquel Ferrer^{2,3}, Gumersindo Gonzalez-Diaz¹, Antonia Lopez-Martinez¹, Noemi Llamas¹, Maravillas Alcazar¹, Lucia Capilla¹, and Antoni Torres^{2,3}



Dias livres de readmissão hospitalar em 1 ano (SHO x DPOC) x Sobrevivência em 1 ano x Dias livres de readmissão hospitalar em 1 ano (DPOC obeso x DPOC não obeso)





VENTILAÇÃO MECÂNICA



VENTILAÇÃO MECÂNICA

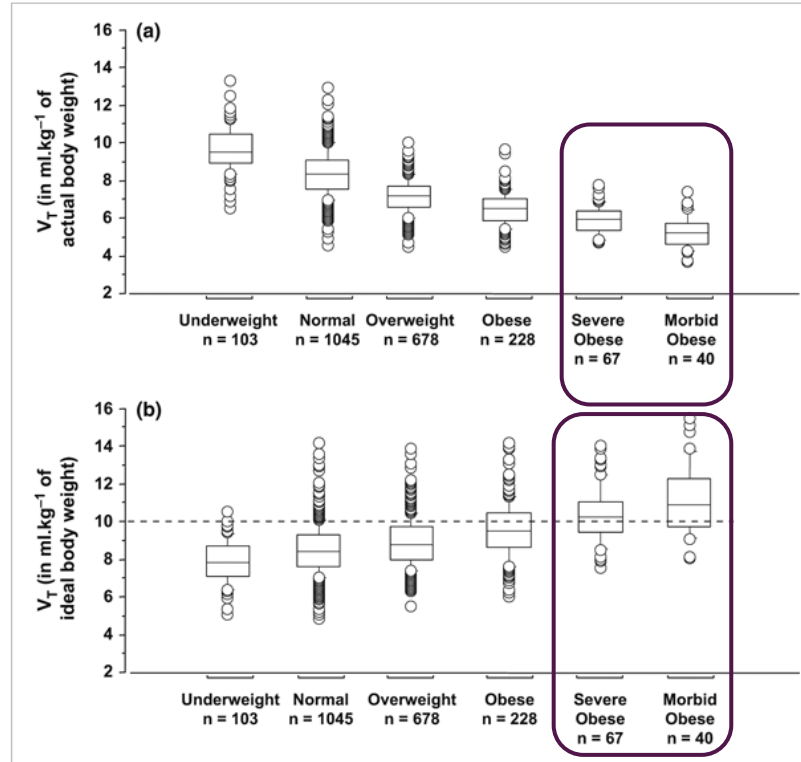
- ❑ Como em não obesos, manter ventilação protetora baseada no IBW com V_c 6-8 mL/kg ou < 6 mL/kg em ARDS
- ❑ Posicionar adequadamente o doente
- ❑ Cuidados com pressão de pico
- ❑ Titulação de PEEP de moderado a alto ≥ 10 cmH₂O desde o começo** → prevenção de derecrutamento x recrutamento de áreas colapsadas
 - limitação intrínseca de fluxo respiratório
 - atentar a efeitos hemodinâmicos, com risco de piora da oxigenação





A multicentre observational study of intra-operative ventilatory management during general anaesthesia: tidal volumes and relation to body weight

S. Jaber, Y. Coisel, G. Chanques, E. Futier, J.-M. Constantin, P. Michelet, M. Beaussier, J.-Y. Lefrant, B.



Positive End-expiratory Pressure Improves Respiratory Function in Obese but not in Normal Subjects during Anesthesia and Paralysis

Paolo Pelosi, M.D.,* Irene Ravagnan, M.D.,† Gabriella Giurati, M.D.,† Mauro Panigada, M.D.,‡
Nicola Bottino, M.D.,‡ Stefano Tredici, M.D.,‡ Giuditta Eccher, M.D.,‡ Luciano Gattinoni, M.D.§

Table 2. Intraabdominal Pressure, Lung Volume, and Elastances of the Total Respiratory System and Lung and Chest Wall at Different PEEP Levels

	0 cm H ₂ O		10 cm H ₂ O	
	Normal	Obese	Normal	Obese
IAP (cm H ₂ O)	9.0 ± 2.4	18.8 ± 7.8**	—	—
EELV (l)	2.154 ± 0.584	0.587 ± 0.175**	2.755 ± 0.595‡	1.030 ± 0.286**‡
E _{rs} (cm H ₂ O/l)	16.39 ± 3.65	26.80 ± 4.20**	15.68 ± 3.23	18.54 ± 3.06‡
E _L (cm H ₂ O/l)	10.28 ± 3.24	17.38 ± 4.50**	9.22 ± 3.01†	12.21 ± 2.48*‡
E _{cw} (cm H ₂ O/l)	6.07 ± 1.43	9.41 ± 3.02**	6.43 ± 1.46	6.33 ± 2.36‡

Table 5. Gas Exchange at Different PEEP Levels

	0 cm H ₂ O		10 cm H ₂ O	
	Normal	Obese	Normal	Obese
V _E (l/min)	9.41 ± 0.77	9.54 ± 0.53	9.41 ± 0.56	9.16 ± 0.68
F _I O ₂ (%)	50 ± 0	50 ± 0	50 ± 0	50 ± 0
PaO ₂ (mmHg)	218.1 ± 47.0	110.2 ± 29.6**	215.3 ± 47.3	130.0 ± 28.0**‡
Δ _(A-a) O ₂ (mmHg)	110.0 ± 45.6	208.5 ± 30.5**	113.3 ± 86.8	187.3 ± 30.2**‡
PaCO ₂ (mmHg)	28.4 ± 3.1	37.8 ± 6.8**	27.8 ± 5.7	39.4 ± 4.9**
pH _a	7.45 ± 0.07	7.38 ± 0.06	7.46 ± 0.09	7.38 ± 0.05*
V _D /V _T (%)	28.7 ± 6.6	47.7 ± 22.2*	27.4 ± 4.5	49.0 ± 15.0**



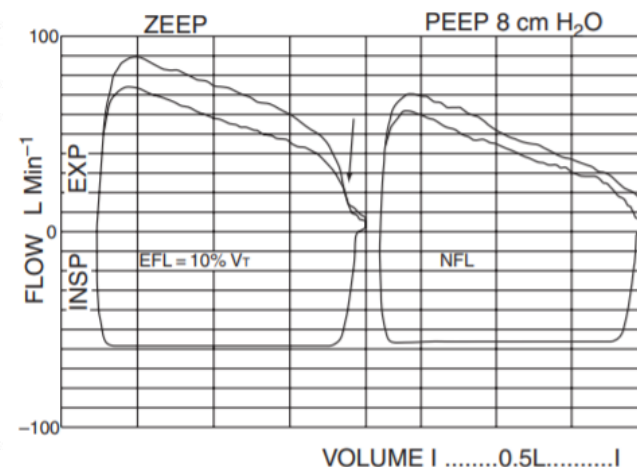
Expiratory flow limitation in morbidly obese postoperative mechanically ventilated patients

A. KOUTSOUKOU, N. KOULOURIS, B. BEKOS, C. SOTIROPOULOU, E. KOSMAS, K. PAPADIMA and C. ROUSSOS

Table 4

Effect of positive end-expiratory pressure (4–16 cm H₂O) on respiratory parameters in seven obese subjects with expiratory flow limitation on zero positive end-expiratory pressure.

	ZEEP	PEEP	P-value
PEEP _{i,st} (cmH ₂ O)	4.5 ± 5.0	1.3 ± 0.9	0.02
P _{plat} (cmH ₂ O)	27.8 ± 4.8	29.6 ± 3.9	NS
R _{max,rs} (cmH ₂ O · l ⁻¹ · s ⁻¹)	12.5 ± 2.5	10.7 ± 1.7	0.03
R _{min,rs} (cmH ₂ O · l ⁻¹ · s ⁻¹)	6.9 ± 1.9	5.9 ± 1.7	NS
ΔR _{rs} (cmH ₂ O · l ⁻¹ · s ⁻¹)	5.6 ± 1.1	4.8 ± 0.8	0.02
ΔFRC (ml)	37 ± 30	174 ± 91	0.02
V _{rec} (ml)		61 ± 13	
Est _{rs} (cmH ₂ O l ⁻¹)	31.1 ± 4.5	27.0 ± 3.5	0.02
ΔP _{aw} /Δt (cmH ₂ O s ⁻¹)	268 ± 62	198 ± 23	0.02
PEEP _i inequality index	0.38 ± 0.19	0.90 ± 0.06	0.02
PaO ₂ (mmHg)	128 ± 47	131 ± 41	NS
PaCO ₂ (mmHg)	36 ± 4	37 ± 5	NS
PaO ₂ /F _{IO2}	219 ± 81	223 ± 61	NS
Δ(A-a)PO ₂	249 ± 74	245 ± 63	NS

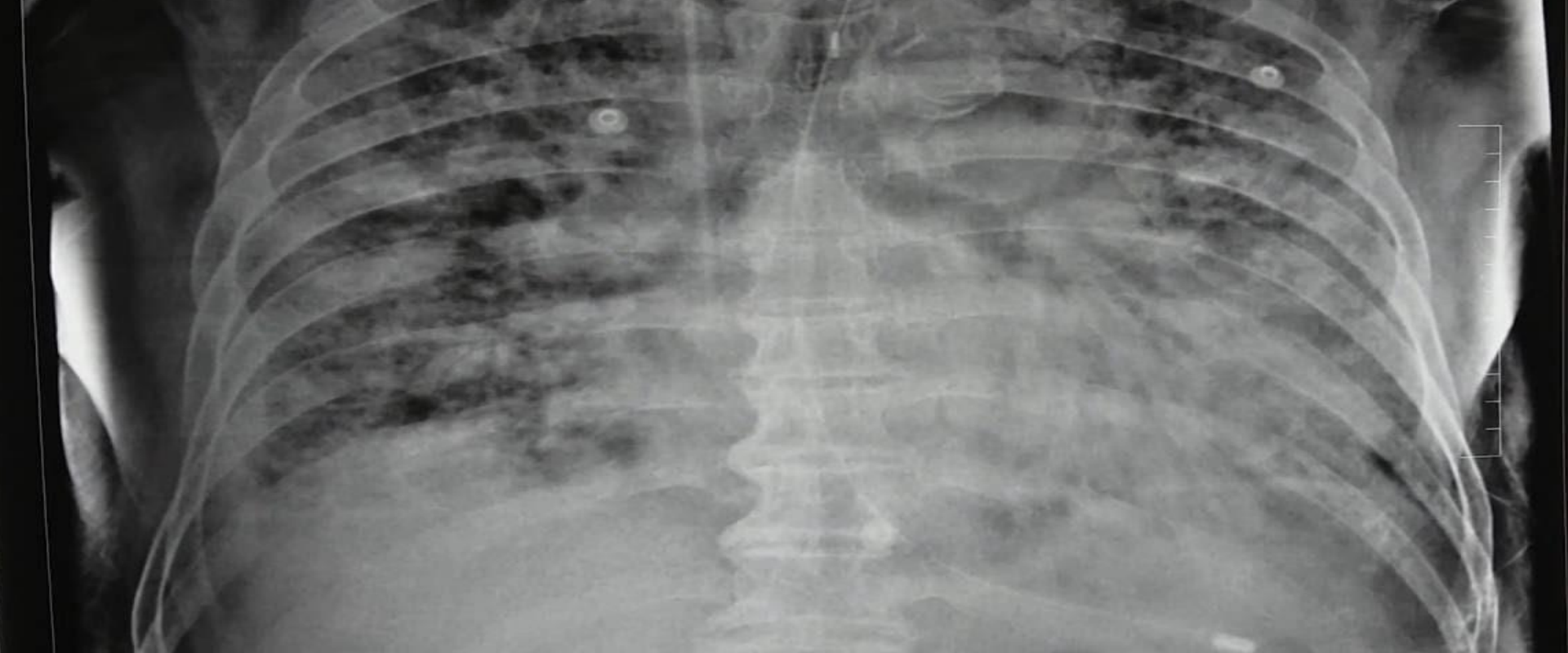


P:15 doentes obesos mórbidos em VM no POI

I: Avaliação de LFE com estratégia de PEEP > encontrado em 10 dos 15 doentes.

C: Aplicação de Zero PEEP = ZEEP





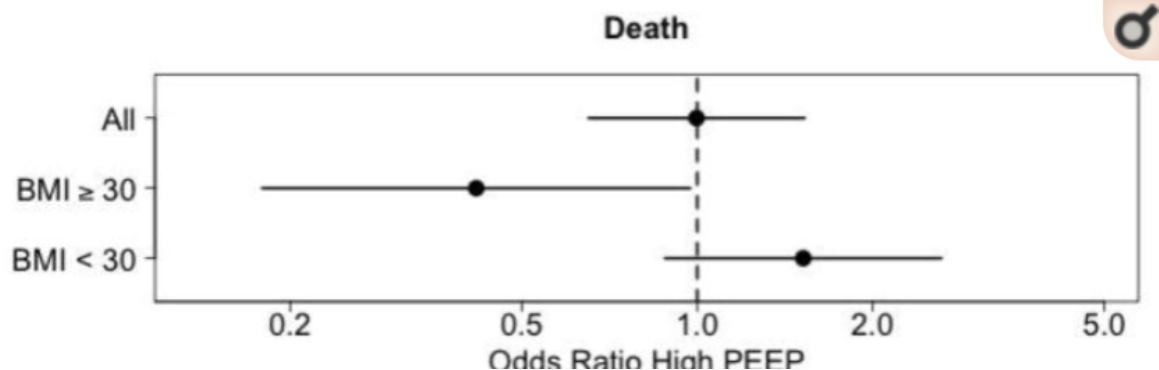
SDRA



High Positive End-Expiratory Pressure Is Associated with Improved Survival in Obese Patients with Acute Respiratory Distress Syndrome.

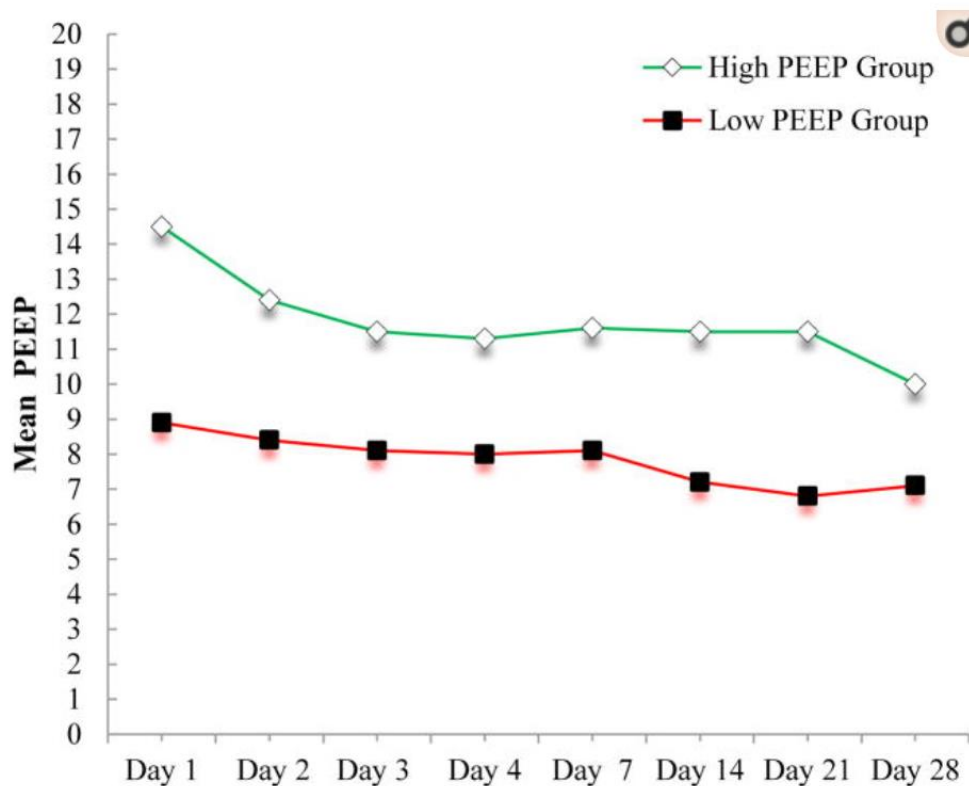
Bime C¹, Fiero M², Lu Z³, Oren E², Berry CE⁴, Parthasarathy S⁴, Garcia JGN⁴.

Outcomes [*]	Non-Obese				Obese			
	High PEEP [£]	Low PEEP [£]	Adjusted OR ^ψ (95% CI)	P value	High PEEP [£]	Low PEEP [£]	Adjusted OR ^ψ (95% CI)	P value
Mortality [§] , %	34	23	1.52 (0.9, 2.6)	0.13	18	32	0.4 (0.18, 0.97)	0.04
Unassisted Breathing [¶] , %	74	76	1.02 (0.6, 1.8)	0.92	80	74	1.52 (0.65, 3.55)	0.34
Barotrauma, %	13	7	1.98 (0.9, 4.4)	0.09	3	6	0.35 (0.06, 2.04)	0.24



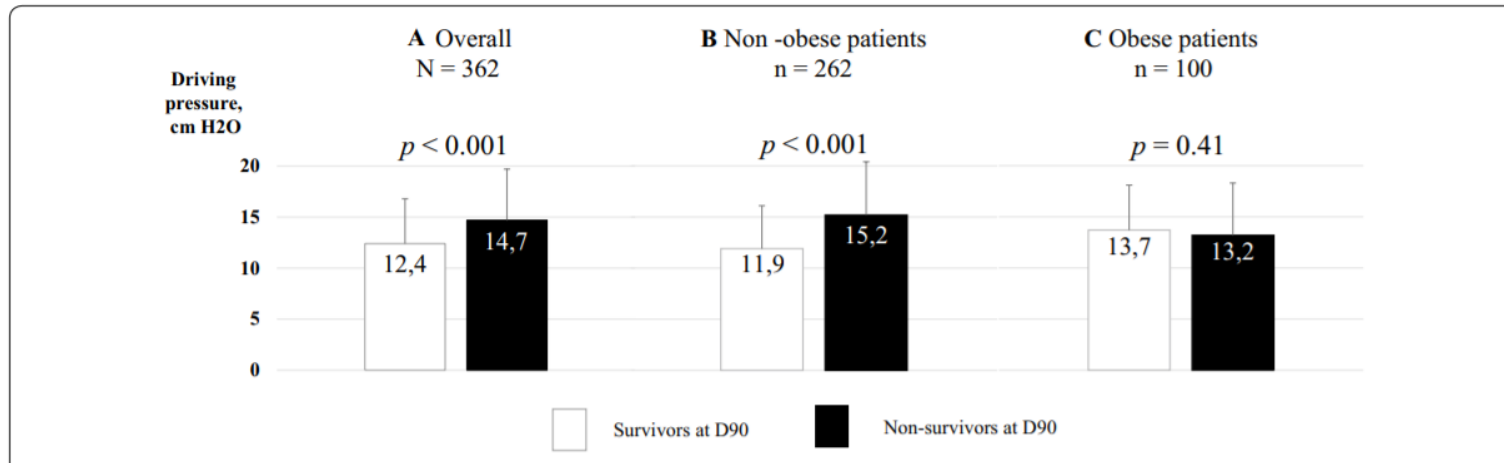
High Positive End-Expiratory Pressure Is Associated with Improved Survival in Obese Patients with Acute Respiratory Distress Syndrome.

Bime C¹, Fiero M², Lu Z³, Oren E², Berry CE⁴, Parthasarathy S⁴, Garcia JGN⁴.



Impact of the driving pressure on mortality in obese and non-obese ARDS patients: a retrospective study of 362 cases

Audrey De Jong¹, Jeanne Cossic², Daniel Verzilli², Clément Monet², Julie Carr², Mathieu Conseil², Marion Monnin², Moussa Cisse², Fouad Belafia², Nicolas Molinari³, Gérald Chanques¹ and Samir Jaber^{1*}



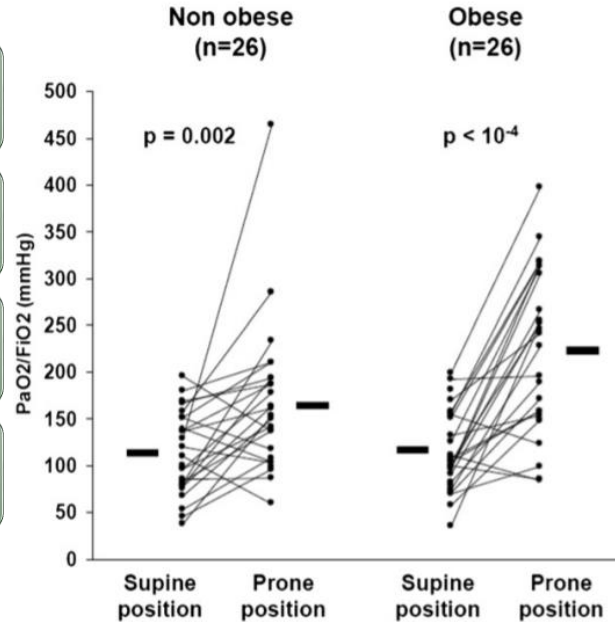
POSIÇÃO PRONA

IMC > 35 kg/m² + P/F < 200 mmHg versus não obesos

Estudo caso-controlle

Avaliação de segurança e complicações como desfecho primário

S: P/F, LOS, tempo de VM, infecções nosocomiais e mortalidade



Chest, 143(6), 1554–1561. doi:10.1378/chest.12-2115



PaO₂/FIO₂ ratio, mm Hg

Before PP	115 ± 42	118 ± 43	113 ± 42	.70
After PP	197 ± 85 ^a	222 ± 84 ^a	174 ± 80 ^a	.03

PaO₂/FIO₂ ratio > 600 mmHg was achieved in 100% of patients in the prone position and in 60% of patients in the supine position.

POSIÇÃO PRONA



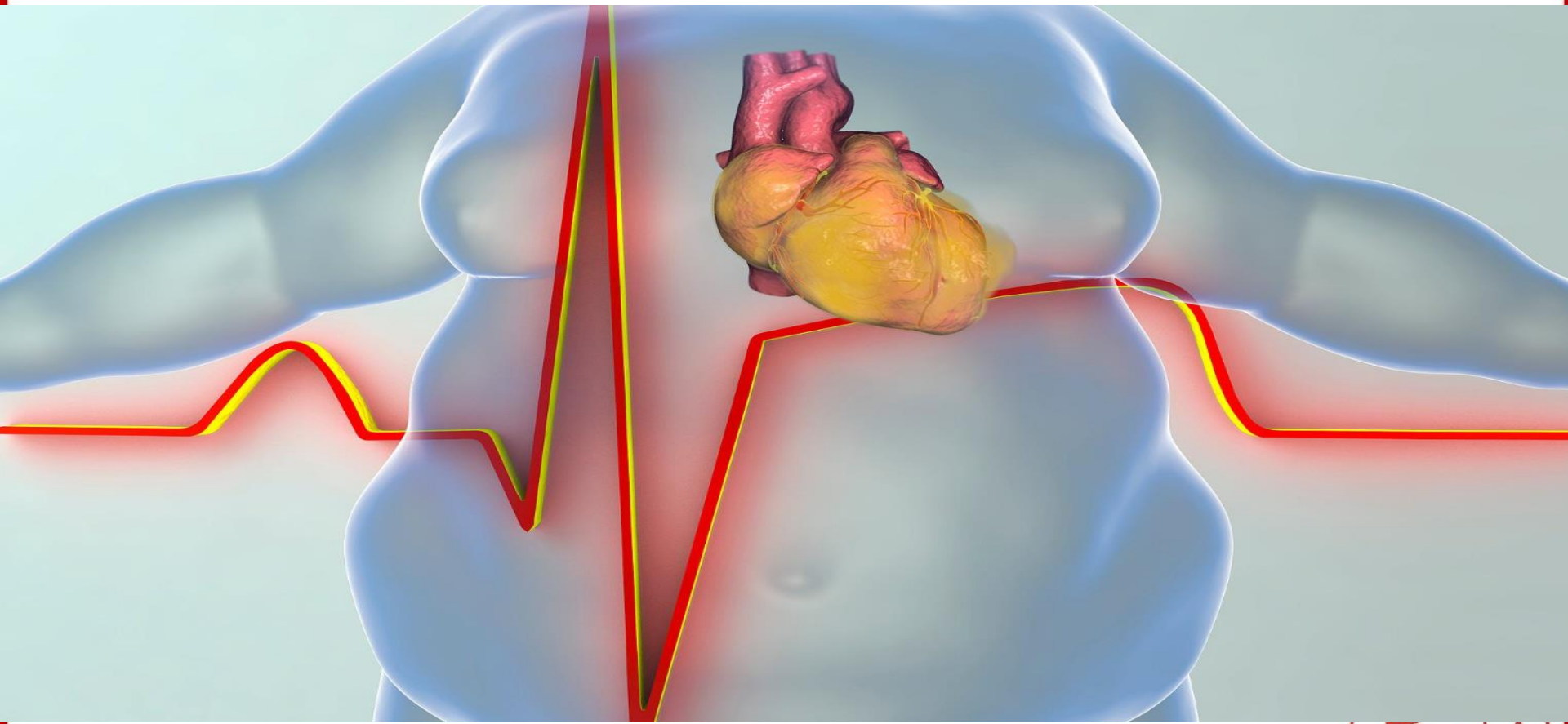
Chest, 143(6), 1554–1561.

POSIÇÃO PRONA

Table 3—Characteristics and Complications of PP in Obese and Nonobese Patients

Variable	All Patients (N = 66)	Obese Patients (n = 33)	Nonobese Patients (n = 33)	P Value
Total number of sessions by patient	2 (1-4)	3 (2-4)	2 (1-3)	.16
Total length of PP by patient, h	22 (10-37)	28 (14-42)	16 (9-30)	.34
Length per session by patient, h	8 (7-11)	9 (6-11)	8 (7-12)	.28
Patients with at least one complication, No. (%)	20 (30)	10 (30)	10 (30)	...
Overall complications, No.	51	25	26	...
Accidental extubation	1	1	0	...
Displacement of endotracheal tube	0	0	0	...
Loss of venous access	1	0	1	...
Loss of arterial access	0	0	0	...
Loss of urinary catheter	0	0	0	...
Loss of gastric tube	0	0	0	...
Forehead pressure ulcers	5	4	1	...
Chin pressure ulcers	10	4	6	...
Other pressure ulcers	6	3	3	...
Total pressure ulcers	16	7	9	...
Facial edema	8	4	4	...
Hemodynamic failure	0	0	0	...
Cardiac arrest	2	1	1	...
Bleeding	1	1	0	...
Vomiting	1	0	1	...





APARELHO CARDIOVASCULAR



ALTERAÇÕES FISIOLÓGICAS

**AUMENTO VOLUME SANGUINEO
CIRCULANTE**

**HIPERTENSÃO ARTERIAL
= HIPERTROFIA CONCÊNTRICA VE
DISFUNÇÃO SISTÓLICA/DIASTOLICA
AUMENTO ATRIAL**

ELEVAÇÃO PIA

**AUMENTO MASSA VD
DISFUNÇÃO VD**

**AUMENTO DA PRÉ-CARGA
= HIPERTROFIA EXCÊNTRICA VE**

**FIBRILAÇÃO ATRIAL
EDEMA PULMONAR**

**QUEDA DO RETORNO VENOSO
CHOQUE “HIPOVOLÊMICO”**

**SENSIBILIDADE A VOLUME
SUSCETIBILIDADE A HIPOTENSÃO**



Influence of body mass index on outcome of the mechanically ventilated patients

A Anzueto,¹ F Frutos-Vivar,² A Esteban,² N Bensalame,¹ D Marks,¹ K Raymondos,³

	Underweight (BMI <18.5 kg/m ²)	Normal (BMI 18.5–24.9 kg/m ²)	Overweight (BMI 25–29.9 kg/m ²)	Obese (BMI 30–39.9 kg/m ²)	Severely obese (BMI >40 kg/m ²)	p-Value
N (% (95% CI))	184 (3.7% (3.2 to 4.3))	1995 (40.1% (38.8 to 41.5))	1781 (35.8% (34.5 to 37.2))	792 (15.9% (14.9 to 16.9))	216 (4.3% (3.8 to 4.9))	
Barotrauma, N (% (95% CI))	8 (4.3% (1.9 to 8.4))	63 (3.1% (2.4 to 4.0))	54 (3.0% (2.3 to 3.9))	27 (3.4% (2.2 to 4.9))	5 (2.3% (0.7 to 5.3))	0.807
Ventilator associated pneumonia, N (% (95% CI))	9 (4.9% (2.3 to 9.1))	68 (3.4% (2.6 to 4.3))	72 (4.0% (3.2 to 5.1))	36 (4.5% (3.2 to 6.2))	12 (5.5% (2.9 to 9.5))	0.380
Sepsis, N (% (95% CI))	22 (11.9% (7.6 to 19.5))	158 (7.9% (6.7 to 9.2))	155 (8.7% (7.4 to 10.1))	50 (6.3% (4.7 to 8.2))	15 (6.9% (3.9 to 11.1))	0.076
Cardiovascular failure, N (% (95% CI))	60 (32.6% (25.9 to 39.9))	620 (31.1% (29.0 to 33.1))	587 (32.9% (30.8 to 35.2))	224 (28.3% (25.1 to 31.5))	63 (29.2% (23.2 to 35.7))	0.179
Respiratory failure, N (% (95% CI))						
With criteria of acute respiratory distress syndrome	11 (5.9% (3.0 to 10.4))	89 (4.5% (3.6 to 5.5))	102 (5.7% (4.7 to 6.9))	55 (6.9% (5.3 to 8.9))	19 (8.8% (5.4 to 13.4))	0.021
Without criteria of acute respiratory distress syndrome	42 (22.8% (16.9 to 29.6))	556 (27.9% (25.9 to 29.9))	557 (31.3% (29.1 to 33.5))	256 (32.3% (29.1 to 35.7))	81 (37.5% (31.0 to 44.3))	0.002
Acute renal failure, N (% (95% CI))	12 (6.5% (3.4 to 11.1))	123 (6.2% (5.1 to 7.3))	150 (8.4% (7.2 to 9.8))	71 (8.9% (7.1 to 11.2))	26 (12.0% (8.0 to 17.1))	0.003
Hepatic failure, N (% (95% CI))	9 (4.9% (2.3 to 9.1))	100 (5.0% (4.1 to 6.1))	83 (4.7 (3.7 to 5.7))	42 (5.3% (3.8 to 7.1))	18 (8.3% (5.0 to 12.8))	0.239
Haematologic failure, N (% (95% CI))	21 (11.4% (7.2 to 16.9))	143 (7.2% (6.1 to 8.4))	127 (7.1% (5.9 to 8.4))	49 (6.2% (4.6 to 8.1))	14 (6.5% (3.6 to 10.6))	0.177



RESSUSCITAÇÃO VOLÊMICA

Intervention ^a mean (SD)		< 18.50 Number = 196	18.50 to 4.99 Number = 1,020	25.0 to 29.99 Number = 816	30.0 to 39.99 Number = 680	≥ 40 Number = 170	P value	Adjusted P value ^b	Adjusted P value ^c
Crystalloid	Number = 1,506 (52.3%)	Number = 96 (49.0)	Number = 537 (52.6)	Number = 439 (53.8)	Number = 346 (50.9)	Number = 88 (51.8)			
	Total (ml)	2,580.3 (1,874.9)	2,690.0 (2,050.2)	2,861.7 (2,445.1)	2,537.4 (2,032.0)	2,602.3 (1,919.1)	0.30	0.76	0.59
	Dose per kg of actual body weight (ml/kg)	55.0 (40.1)	43.2 (33.4)	37.1 (30.8)	27.7 (22.0)	21.4 (16.8)	< 0.001	< 0.001	< 0.001
Colloid	Number = 1,506 (52.3%)	Number = 96 (49.0)	Number = 537 (52.6)	Number = 439 (53.8)	Number = 346 (50.9)	Number = 88 (51.8)			
	Total (ml)	656.8 (864.8)	509.8 (781.0)	593.5 (788.7)	705.5 (2,824.5)	538.9 (939.8)	0.44	0.79	0.31
	Dose per kg of actual body weight (ml/kg)	14.3 (19.3)	8.2 (12.7)	7.7 (10.2)	8.5 (44.2)	4.3 (7.2)	0.06	0.02	< 0.001

Arabi et al. Critical Care 2013, 17:R72



RESSUSCITAÇÃO VOLÊMICA

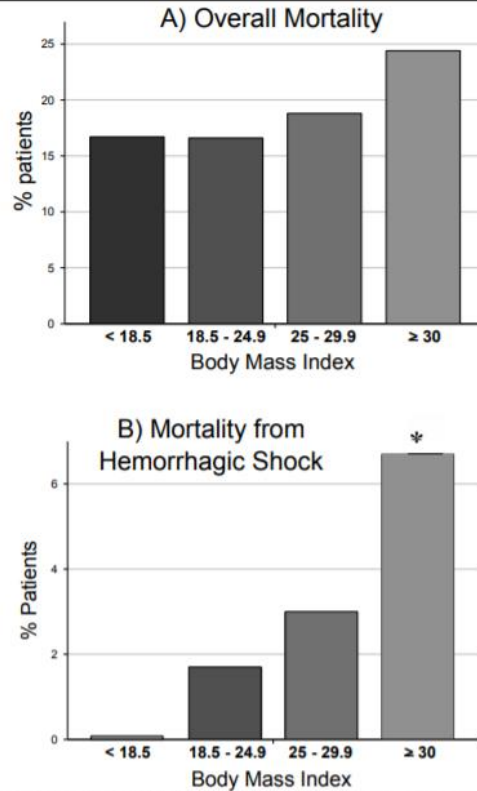
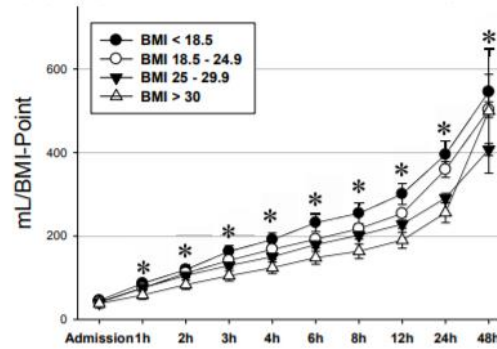


Figure 1 Mortality as a function of body mass index. (A) Overall mortality and (B) mortality from hemorrhagic shock. *Significant differences between patients with body mass index of 18.5 to 24.9 compared with patients with BMI > 30.

(A) Crystalloid Volume corrected per BMI



(B) Colloid Volume corrected per BMI

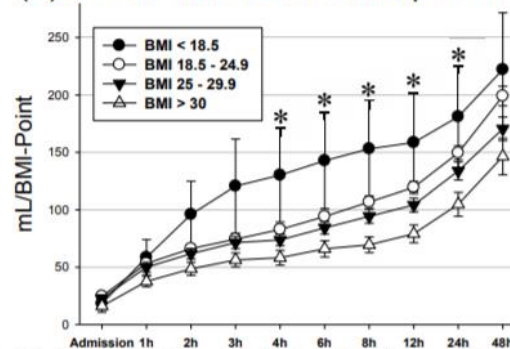


Figure 3 Crystalloid and colloid volume per body mass index point. Amount of (A) crystalloid volume and (B) colloid volume corrected for body mass index (BMI) point (total amount/number of BMI points) per BMI group. Error bars indicate the standard error of the mean. *Significant differences between patients with BMI of 18.5 to 24.9 compared with patients with BMI > 30.

Arabi et al. Critical Care 2013, 17:R72





NUTRIÇÃO



	BASELINE	ESTRESSE
Glicose	++	++++
GH	-	=
Norepinefrina	++	--
Cetonas e AGL	++	++
Insulina*	++	-
Cortisol	++	=

*** Supressão da mobilização de estoques de lipídios das reservas e desvio da gliconeogênese para quebra de proteínas**



CALCULO ENERGÉTICO

- ❑ Não é simples determinação de necessidades calóricas
- ❑ Melhor forma por calorimetria indireta
- ❑ Uso de Harris-Beneficit não é acurado (74% de erro usando ABW e 36% usando ADjBW) tendendo invariavelmente à subestimação



Table 3 Summary of the ESPEN and ASPEN guidelines for adjustment of nutritional therapy in critically ill obese patients

Suggestions	ESPEN guidelines	ASPEN guidelines
For calculating the energy target if measurement of REE is not possible		
In general	20–25 kcal/kg actual BW/day Below 70% of REE should be given during 'early' acute phase	25–30 kcal/kg actual BW/day
In obese	Same as above, but calculated according to adjusted BW ^a If REE measured, set target to 80–100% of REE after the early acute phase (within days 3–7)	11–14 kcal/kg actual BW/day if BMI 30–50 kg/m ² 22–25 kcal/kg ideal BW ^b /day if BMI > 50 kg/m ² If REE measured, set target to 65–70% of REE
For calculating protein target		
In general	1.3 g/kg actual BW/day	1.2–2.0 g/kg actual BW/day
In obese	Same as above, but calculated with adjusted BW ^a	2.0–2.5 g/kg ideal BW ^b /day



Suggestions	ESPEN guidelines	ASPEN guidelines
Example 1: male 120 kg, 185 cm $\geq$ BMI = 35.1 kg/m ² Ideal BW ^b = 77 kg ^b and adjusted BW ^a 86–88 kg		
Energy target	Calculated with adjusted BW ^a 25 kcal $\times$ 86–88 kg Target = 2150–2200 kcal/day	Calculated with actual BW 14 kcal $\times$ 120 kg Target = 1680 kcal/day
Protein target	Calculated with adjusted BW ^a 1.3 g $\times$ 92–96 kg ^a Target = 120–125 g/day	Calculated with ideal BW ^b 2.0–2.5 g $\times$ 77 kg ^b Target = 154–193 g/day
Example 2: female 140 kg, 165 cm $\geq$ BMI = 51.5 kg/m ² Ideal BW ^b : 53 kg and adjusted BW ^a = 70–75 kg		
Energy target	Calculated with adjusted BW ^a 25 kcal $\times$ 70–75 kg Target = 1750–1875 kcal/day	Calculated with ideal BW ^b 25 kcal $\times$ 53 kg Target = 1325 kcal/day
Protein target	Calculated with adjusted BW ^a 1.3 g $\times$ 70–75 kg Target = 91–98 g/day	Calculated with ideal BW ^b 2.0–2.5 g $\times$ 53 kg Target = 106–133 g/day



MONITORIZAÇÃO

Suggestions	ESPEN guidelines	ASPEN guidelines
For adjustment of nutritional therapy according to serum markers ^c		
Glucose	Below 10 mmol/l (180 g/l) Consider lowering carbohydrate administration when > 6 U insulin/h is needed for > 24 h	Below 10 mmol/l (180 g/l)
Urea	Consider lowering protein administration if > 30 mmol/l: Probably only justified if protein administration > 1.5 g/kg BW/day	–
Triglycerids	Investigate and consider lowering fat administration if > 5.6 mmol/l	–
Examples for calculating energy and protein targets in obese ^d		



EXEMPLO 3

Nutrition method	Macronutrients	Rate	Calorie and protein intakes
Parenteral Nutrition: Concentration-based D10W amino acids 8% lipids 2%	D10 W amino acids 8% lipids 2%	75 mL/hr	1638 kcals or 24kcal/kg IBW 144 g protein or 2.1 g protein/kg IBW
Parenteral Nutrition: ASPEN Safe Practices	Dextrose 200 g, amino acids 140 g Lipids 40 g/d	75 mL/hr	1640 kcals or 24kcal/kg IBW 140 g or 2.1g protein/kg IBW
Enteral formula type	Macronutrient concentration		
Ultra-high protein	1 kcal/mL 92 g protein/L	70 mL/h	1680 kcals or 25 kcal/kg IBW 155 g or 2.3 g protein/kg IBW
Ultra-high protein	1 kcal/mL 92 g protein/L	65 mL/h	1560 kcals or 23 kcal/kg IBW 144 g or 2.1 g protein/kg IBW
High protein	1 kcal/mL 64 g protein/L	55 mL/h	1720 kcals or 25 kcal/kg IBW 144 g or 2.1 g protein/kg IBW
Protein boluses	Protein* 30 g twice daily		
High protein	1 kcal/mL 64 g protein/L	50 mL/h	1600 kcals or 24 kcal/kg IBW 137 g or 2.0 g protein/kg IBW
Protein boluses	Protein* 30 g twice daily		

ASPEN, American Society for Parenteral and Enteral Nutrition; BMI, body mass index; IBW, ideal body weight

Patient case: A 40-y old male, weight 110 kg, height 173 cm, IBW 68 kg, BMI 36.8 kg/m², normal renal and hepatic dysfunction, without hyperglycemia, hypertriacylglycerol-mia, or a history of diabetes mellitus





APARELHO URINÁRIO



CLASSIFICAÇÃO LRA

- ❑ Existe uma limitação dos critérios atualmente utilizados para definição de LRA base no débito urinário - tendendo a classificar erroneamente doentes com TFGs abaixo das verdadeiras
- ❑ É sugerido utilizar os cutoffs baseados na creatinina sérica



**Comparative Performance of the CKD Epidemiology
Collaboration (CKD-EPI) and the Modification of Diet in Renal
Disease (MDRD) Study Equations for Estimating GFR Levels
Above 60 mL/min/1.73 m²**

Lesley A. Stevens, MD MS¹, Christopher H. Schmid, PhD¹, Tom Greene, PhD², Yaping

Group		Equation	Overall			eGFR < 60			eGFR 60 - 89			eGFR > 90		
			No.	Bias	%Bias	No.	Bias	%Bias	No.	Bias	%Bias	No.	Bias	%Bias
BMI (kg/m ²)	<20	}CKD-EPI	249	-3.5	-6.8	120	-1.7	-6.3	73	-1.2	-1.9	56	-16.6	-18.1
	20-25		1357	2.2	4.2	589	1.7	4.9	341	4.5	5.7	427	1.5	1.5
	26-30		1339	3.2	6.3	643	2.3	7.1	350	4.9	6.7	346	5.2	4.4
	>30		951	2.1	4.4	550	2.1	6.1	226	3.9	5.1	175	-0.9	-0.7
	<20	}MDRD 	249	0.6	1.0	139	0.1	0.3	74	7.4	8.9	36	-13.7	-14.0
	20-25		1357	5.9	10.7	656	4.1	10.6	408	12.2	14.2	293	6.8	6.4
	26-30		1339	6.7	12.0	705	3.7	9.9	412	13.4	15.4	222	9.9	8.1
	>30		951	4.4	9.6	600	3.2	9.6	230	10.3	12.4	121	6.3	5.3



African American race, obesity, and blood product transfusion are risk factors for acute kidney injury in critically ill trauma patients

Michael G. S. Shashaty, M.D., M.S.C.E.^{1,2}, Nuala J. Meyer, M.D., M.S.¹, A. Russell Localio,

[J Crit Care](#). 2012 Oct;27(5):496-504

- **IMC 30 kg/m ou mais(OR, 4.72 versus IMC normal; 95% CI, 2.59-8.61; P < .001)**

Intensive Care Med (2010) 36:1221–1228
DOI 10.1007/s00134-010-1844-2

ORIGINAL

Wilfred Druml
Barbara Metnitz
Eva Schaden
Peter Bauer
Philipp G. H. Metnitz

Impact of body mass on incidence and prognosis of acute kidney injury requiring renal replacement therapy

- **Aumento de LRA desde abaixo do peso, normal (5.4%) a obeso mórbido (11.8%)**

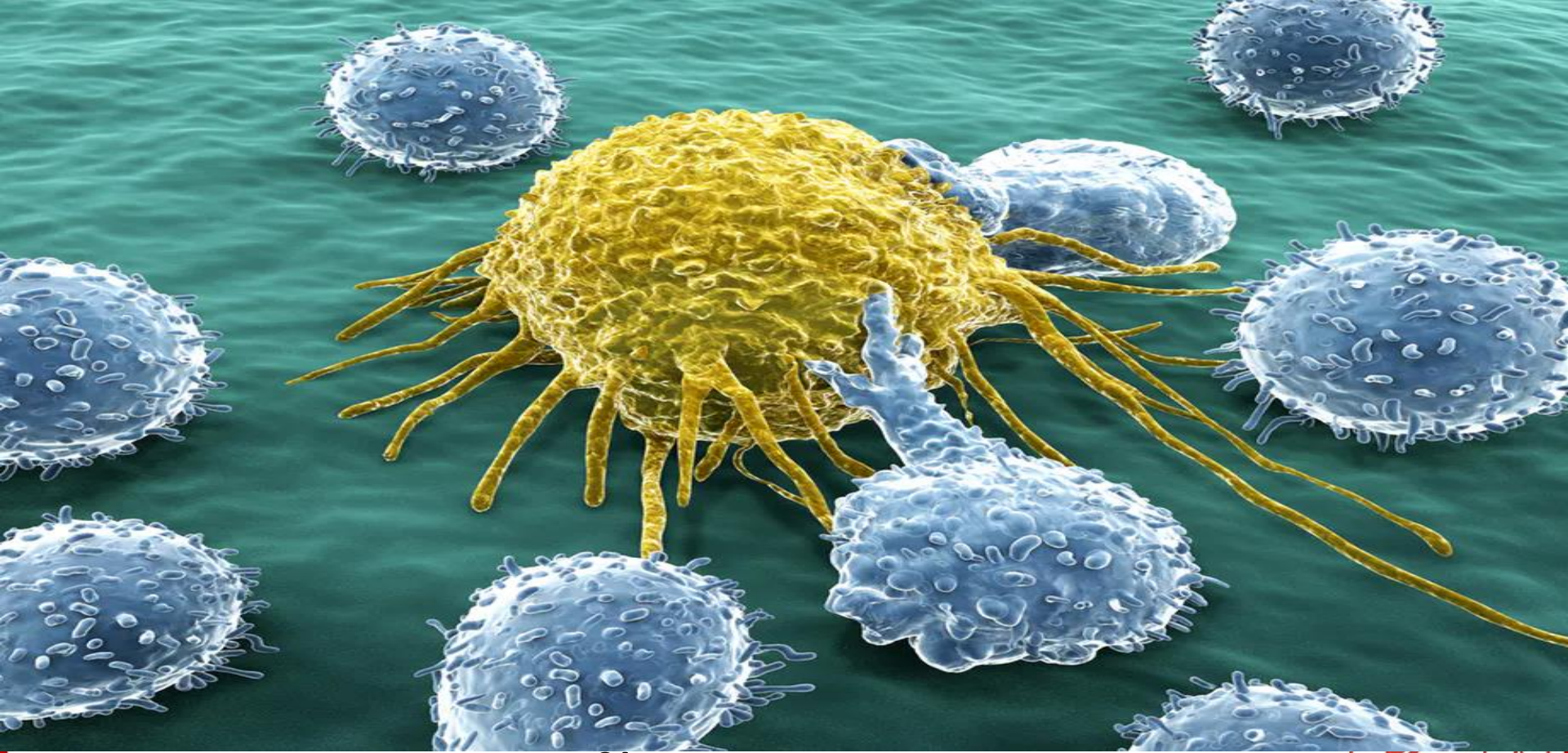
Body Mass Index and Acute Kidney Injury in the Acute Respiratory Distress Syndrome

Graciela J Soto, MD, MS^a, Angela J Frank, MD^c, David C Christiani, MD, MPH^{b,c}, and Michelle Ng Gong, MD, MS^{a,d}

Crit Care Med. 2012 September ; 40(9): 2601–2608

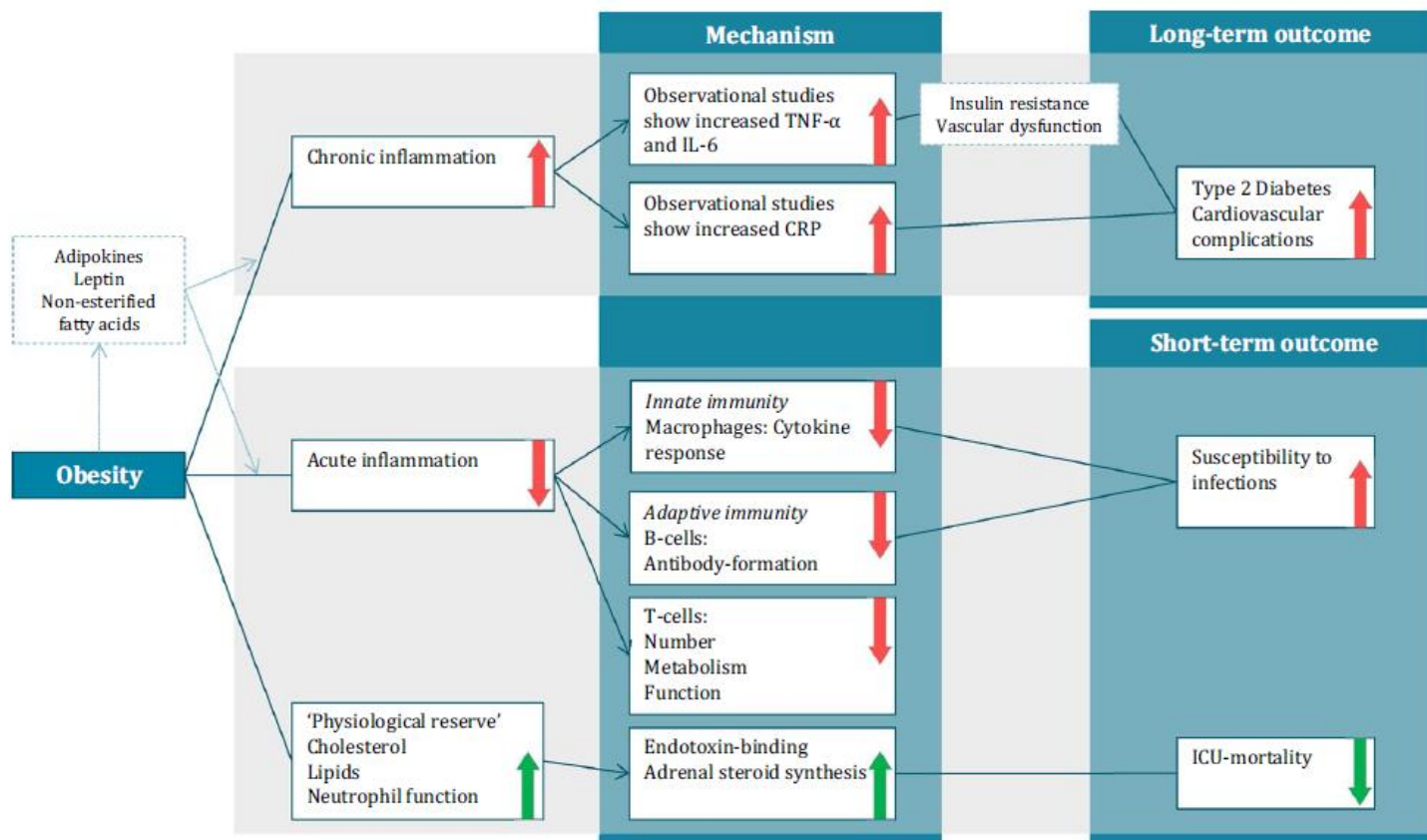
- **IMC significativamente associada a LRA(ORadj 1.20 por aumento de 5 kg/m² no IMC, (95%CI 1.07–1.33).**





INFLAMAÇÃO E IMUNIDADE





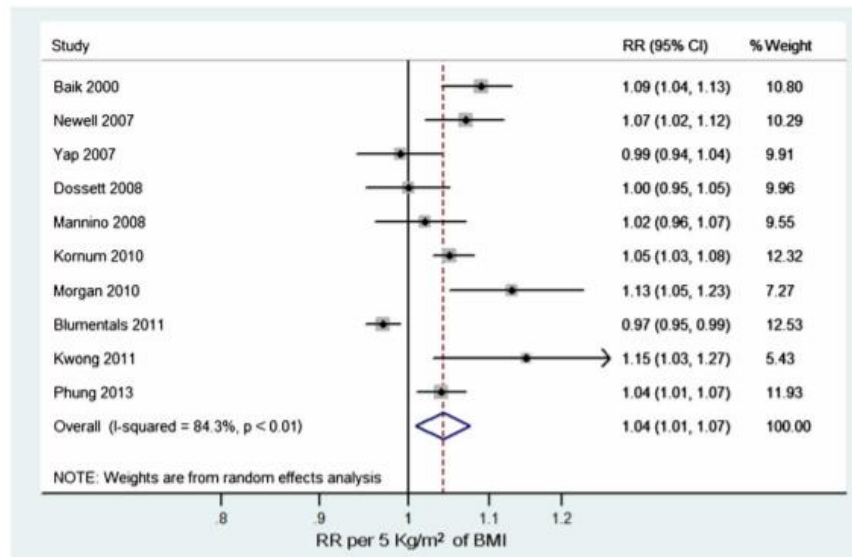


Figure 2 Relative risks of pneumonia risk per 5 kg/m² increase in body mass index. CI: indicates confidence interval; and RR: risk ratio.





FARMACOCINÉTICA

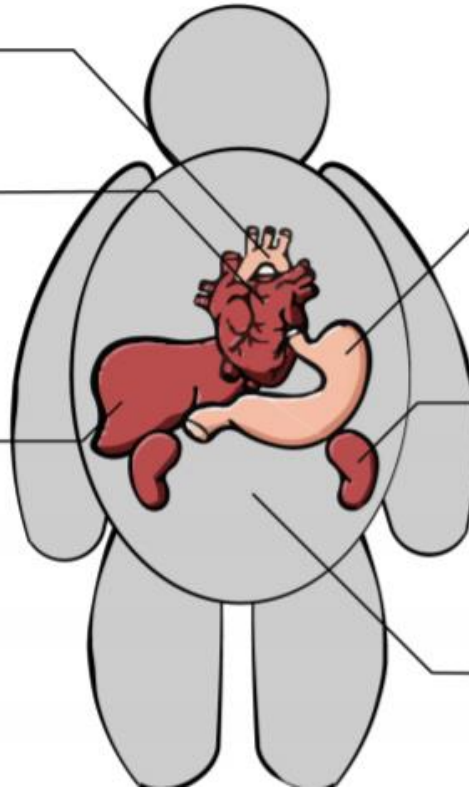


ALTERAÇÕES FISIOLÓGICAS

Serum albumin	↔
Serum AAG	↔ or ↑
Protein binding	↔

Cardiac output	↑
----------------	---

Liver blood flow	↑ or ↔
CYP3A4 activity	↓
CYP2E1 activity	↑
Other CYP activity	↕ or ↔
Phase II reactions	↑
Drug transporter activity	↕
CYP-mediated clearance	↔
or ↕ **	
Glucuronidation	↔ or ↑**
Sulphatation	↔ or ↑***
Glucuronide elimination	↓**



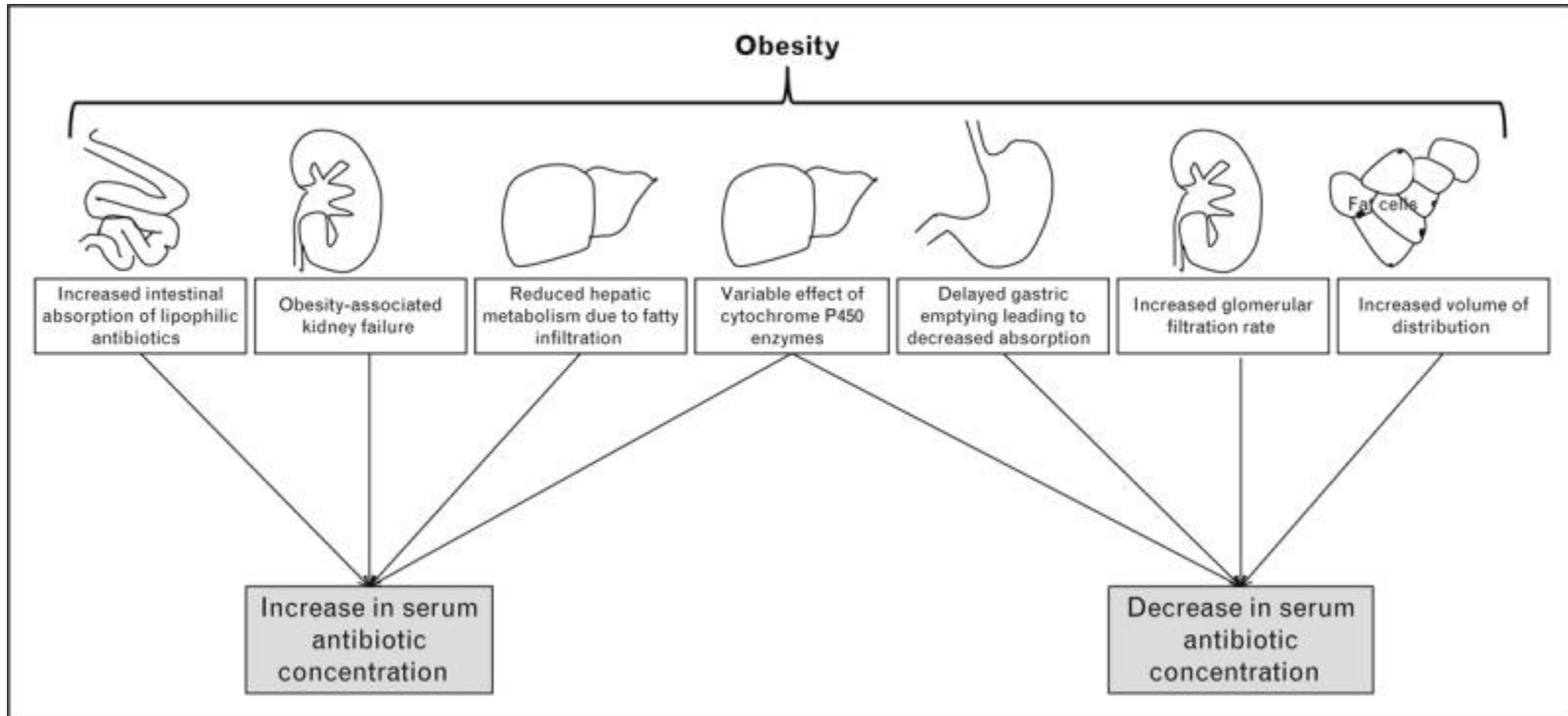
Gastric emptying time	↑
Gut permeability	↑
Gut enzym activity	↕
Oral bioavailability	↔
or ↑ (morbidly obese)	
SC bioavailability	↔

GFR	initially ↑
	later ↓
Tubular secretion	↑
Renal clearance	initially ↑
	later ↓

Adipose tissue	↑
Lean:fat ratio	↓
Volume of distribution / TBW *	↕
Tissue penetration	↓



ALTERAÇÕES FISIOLÓGICAS





ANTIMICROBIANOS

DROGAS HIDROFILICAS

DROGAS LIPOFILICAS

FARMACOCINÉTICA

- Baixo volume de distribuição*
- Clareamento renal
- Menor penetração celular e tissular

- Alto incremento volume de distribuição
- Clareamento hepático
- Maior penetração celular e tissular

MUDANÇAS NA OBESIDADE

- Pouco impacto no volume de distribuição
- Clearance renal aumentado

- Aumento no volume de distribuição
- Clearance hepático variável

DOSES NO OBESO

- Conforme IBW ou AdjBW

- Peso corporal total

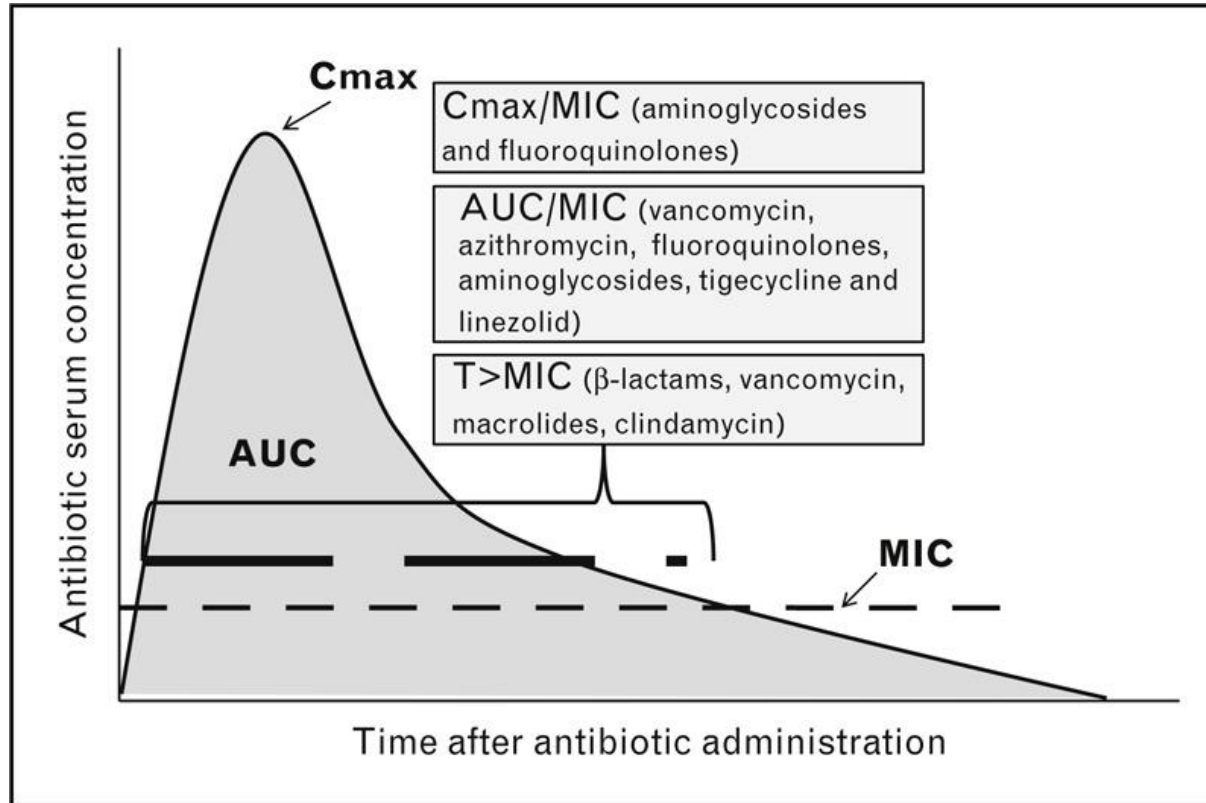
EXEMPLOS

- Betalactamicos
- Aminoglicosídeos
- Vancomicina
- Colistina

- Fluoroquinolonas
- Macrolídeos
- Tigeciclina



ANTIMICROBIANOS



REGIME DE AJUSTE DE DOSES

- ❑ Sedativos (benzodiazepínicos e propofol)
 - *Dose de ataque no peso total e manutenção com base em IBW e titulação conforme resposta clínica*



	Dexmedetomidine	Propofol	Midazolam	Lorazepam
Mechanism of action	α_2 -agonist	GABA _A agonist (with other effects)	GABA _A agonist	GABA _A agonist
Onset	5–10 min	1–2 min	3–5 min	15–20 min
Half-life	Up to 3 h, duration 60–120 min	26–32 h	2–6 h, duration < 2 h (dose dependent)	8–15 h
Metabolism	Hepatic, N-glucuronidation and N-methylation, CYP2A6	Hepatic, CYP2B6	Hepatic CYP3A4	Hepatic, glucuronidation
Active metabolite	None	None	Yes, 1-hydroxymidazolam (accumulates with prolonged infusion, renally excreted)	None
Adverse effects	Bradycardia, hypotension, xerostomia	Hypotension, respiratory depression, hypertriglyceridemia, pain on injection, pancreatitis, PRIS	Respiratory depression	Respiratory depression, propylene glycol-related acidosis, renal failure
Dose	0.2–1.5 $\mu\text{g/kg/h}$	5–50 $\mu\text{g/kg/min}$ (doses greater than 80 $\mu\text{g/kg/min}$ increase risk of HTG and PRIS)	0.5–2 mg	0.5–2 mg
Pharmacokinetics in obese patients	No studies	No significant change	Increased volume of distribution, increased terminal elimination half-life, prolonged duration of action with infusion	Increased volume of distribution, increased terminal elimination half-life, prolonged duration of action with infusion
Dosage in obese patients	No specific dosing recommendations. Typically based on TBW	Based on TBW	Loading dose based on TBW, maintenance dosing based on IBW	Loading dose based on TBW, maintenance dosing based on IBW

HTG hypertriglyceridemia, PRIS propofol infusion syndrome, TBW total body weight, IBW ideal body weight

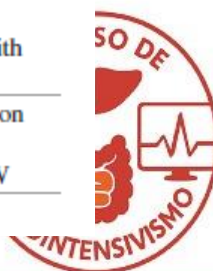


Table 12.1 Analgesics

	Fentanyl	Morphine	Hydromorphone	Remifentanyl
Mechanism of action	μ -opioid agonist	μ -opioid agonist	μ -opioid agonist	μ -opioid agonist
Onset	1–2 min	5–10 min	5–10 min	1–3 min
Elimination half-life	2–4 h	3–4 h	2–3 h	3–10 min
Metabolic pathway	N-dealkylation CYP3A4/5 substrate	Hepatic, glucuronidation	Hepatic, glucuronidation	Hydrolysis by plasma esterases
Active metabolite	None	6- and 3- glucuronide metabolite (renally excreted)	None	None
Intermittent dosing	0.35–0.5 μ g/ kg IV q0.5–1 h	2–4 mg IV q1–2 h	0.2–0.6 mg IV q1–2 h	None
Continuous infusion	25–200 μ g/h	2–30 mg/h	0.5–3 mg/h	1.5 μ g/kg IV loading dose; then 0.5–15 μ g/ kg/h
Side effects	Less hypotension; accumulation with hepatic impairment	Accumulation with hepatic/ renal impairment, histamine release	Accumulation with hepatic/ renal impairment	No accumulation with organ dysfunction
Pharmacokinetics in obese patients	No significant change	No significant change	No significant change	No significant change
Dosage in obese patients	Based on LBW	Based on IBW	Based on IBW	Based on LBW

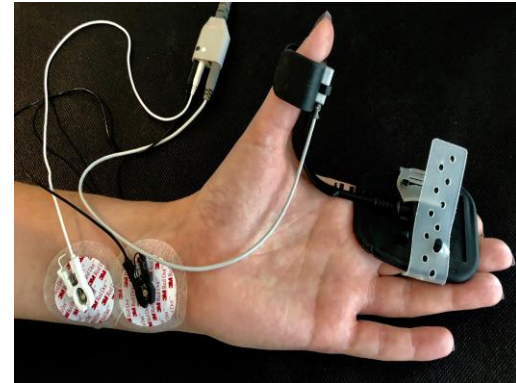
LBW lean body weight, IBW ideal body weight



REGIME DE AJUSTE DE DOSES

- NMBA + reversores
 - *Não-despolarizantes: ATK + manutenção por IBW*
 - *Succinicolina: basear em TBW até 140 mg*
 - *Sugammadex: basear no TBW*

... Lembrar estimulação de nervo periférico



REGIME DE AJUSTE DE DOSES

❑ Anticoagulantes

HNF: dose de ATK + inicial conforme ABW + controles por TTPA

Enoxaparina: utilizar dose-limite até possibilidade de dosagem de anti-Xa

- *Doses maiores em profilaxia de bariátrica*
- *Variação em espectro de dosagem sugerida conforme indicação*



REGIME DE AJUSTE DE DOSES

- Anticoagulantes

ACO oral: cinética errática em IMC > 40; podem ser tentados NOACs com monitorização de efeito de pico com antiXa



SUMÁRIO E CONCLUSÕES

- ☐ Doentes obesos impõem desafios aos cuidados, desde logística multiprofissional, porém geralmente performam bem na fase aguda das afecções
- ☐ Evitar overfeeding parece ser adequado, com dietas hipocalóricas e hiperproteicas
- ☐ Manejo de via aérea deve levar em consideração preditores conhecidos, que geralmente estão elevados no obeso
- ☐ Doentes obesos podem ser ventilados com estratégias protetoras de maneira similar a outros grupos, contudo podem necessitar de valores de PEEP intermediários
- ☐ Evitar cálculo de doses e VM sem conhecimento prévio do IBW e ABW!!



Obrigado!

Dúvidas?

Encaminhar para gastrointensiva@gmail.com

